The Cost of Inclusion: Decomposing Bidder Exclusion in Public Procurement*

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Abstract

SME set-asides are deployed in over a hundred jurisdictions worldwide and made a procurement default by Brazilian Law 14,133/2021. Their welfare turns on whether the price cost flows through the change in admissible bidders or through how each remaining bidder bids—a question price regressions cannot answer. I exploit a March 2018 administrative reversal that extended an SME-only rule to medical supplies on São Paulo's centralized state procurement platform, which intermediates 4.8 million item-level transactions over the structural sample window. A difference-in-differences against 76 never-treated product groups estimates a 10–11 percent rise in absolute winning prices, with SME participation roughly doubling. An asymmetric IPV model identified from Pregão drop-out bids decomposes the price response under observed equilibrium entry. The within-auction component accounts for 74 percent of the simulated effect in non-pharmaceuticals and 73 percent in pharmaceuticals; endogenous SME entry attenuates a 50–60 percent larger latent shock. At $\lambda = 0.30$, the implied welfare cost reaches 28.7 percent of the open-regime baseline price in non-pharmaceuticals and 47.0 percent in pharmaceuticals, scaling to an annual welfare loss of R$ 55–128 million (US$ 16–37 million) on Group-65 procurement alone. A 10 percent SME price preference welfare-dominates

* *This version: April 2026. I thank Paulo Furquim de Azevedo for early discussions and the BEC/SEFAZ-SP team for data access. Replication code and data dictionary are available at the project repository. All remaining errors are my own.*

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the set-aside in thick standardized markets at near-zero fiscal cost; the comparison reverses in thin pharmaceutical markets under strict primitive invariance. The conditional ranking is itself the policy-design statement.
Keywords: public procurement, SME set-asides, asymmetric auctions, structural estimation, welfare, marginal cost of public funds

JEL MSC: H32, H57, L26, L53, D44

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1. Introduction

Procurement set-asides for small firms are among the most widely deployed government-procurement instruments worldwide (Bosio et al., 2022). They were designed on a textbook intuition: bringing more bidders to a protected segment should reduce procurement prices. The intuition has not survived contact with the data. A decade of empirical evidence places the price cost of bidder-exclusion rules in the ten-to-fifteen percent range (Marion, 2007; Krasnokutskaya, 2011; Athey et al., 2013; Nakabayashi, 2013; Hyytinen et al., 2018). What price regressions cannot deliver is the channel: whether the markup runs through the change in *who* can bid, or through the change in *how* each remaining bidder bids. The welfare ranking of bidder exclusion against alternative preference instruments turns on which mechanism dominates, and the policy fix differs by an order of magnitude depending on the answer. If the markup is extensive, the prescription is to widen the protected roster. If it is intensive, no roster expansion will close the gap; the only lever is to shrink the rule. On São Paulo's medical-supplies procurement alone, the structural exercise of this paper prices the rule at R$ 55–128 million (US$ 16–37 million) per year of welfare loss—a magnitude this paper traces to a single mechanism.

A March 2018 administrative reversal in São Paulo state, Brazil, supplies the natural experiment. The state extended an existing SME-only procurement rule to medical and hospital supplies—the last major exception—on the country's largest centralized state-

government procurement platform, the Bolsa Eletrônica de Compras (BEC). The reversal was a legal opinion internal to the state prosecutor's office, defended on isonomy grounds and uncorrelated with prices, competition, or supplier complaints in the affected product group; the timing was bureaucratic. A difference-in-differences against 76 never-treated product groups, reported as design validation in Appendix Appendix I, estimates that absolute winning prices rose 10–11 percent on the same window used by the structural exercise, with SME participation roughly doubling. The DiD coefficient fixes the average effect on observed log prices; it does not say which mechanism produced the markup or what the rule costs in welfare terms. The structural exercise of this paper recovers both, anchored to the same magnitude.

Three magnitudes appear in this paper and should not be confused. The DiD coefficient (10–11 percent) is the average effect on observed log winning prices, item and month fixed effects partialled out. The structural counterfactual (+0.259 in non-pharma, +0.308 in pharma) is the simulated mean of the second-order statistic under each counterfactual entry profile, scaled by the reference price; in p_{S_1} units it is three to four times the DiD coefficient because the two objects average over different primitives. The welfare cost (28.7 percent of p_{S_1} in non-pharma, 47.0 percent in pharma at $\lambda = 0.30$) adds the marginal-cost-of-public-funds distortion on top of the allocative wedge. Sections 6 and 7 develop these in turn; Appendix Appendix I reports the DiD.

This paper provides that structure. I specify an asymmetric independent-private-values reading of the platform's Pregão English-reverse auction format. Bidders draw costs from type-specific distributions F_c^{SME} and $F_c^{\neg}$ and compete under a descending clock; losers exit at cost and the last firm supplies the contract. Drop-out prices of losers therefore point-identify F_c^k under Haile and Tamer (2003), without inverting a bid function or imposing parametric structure on bidding. An auction-level common scale is shrunk out following Krasnokutskaya (2011); winners are treated as left-censored at the second-lowest

exit and recovered by a Turnbull NPMLE robustness. Entry is handled in the Athey et al. (2011) spirit: pre- and post-period bidder counts are taken as equilibrium arrival rates, and the main specification allows the post-policy SME cost distribution to differ from its pre-policy counterpart as part of the equilibrium-selection object. The exercise is conducted under observed equilibrium entry; it is not a fully solved nested entry-bidding equilibrium, and the decomposition that follows is a counterfactual-simulation accounting under the modeling assumptions stated.

Three findings organize the paper.

(i) A 10 percent price preference welfare-dominates the set-aside at near-zero fiscal cost. In thick standardized markets the simulated price shifts by -0.004 of the reference price under the preference, against $+0.259$ under the set-aside. The dominance holds under both the main equilibrium-selection and the strict-primitive-invariance treatment of the post-policy SME pool. Scaled to the post-policy run-rate of Group-65 procurement on São Paulo's R$ 13 billion centralized platform, the implied annual welfare loss of the set-aside is on the order of R$ 55–128 million (US$ 16–37 million).

In thin heterogeneous pharmaceutical markets the comparison bifurcates. The preference dominates the set-aside under the main equilibrium-selection reading; under strict primitive invariance the ranking flips, with the implicit welfare weight required to prefer the set-aside falling to 0.7. The bifurcation is the diagnostic: it pins down where the modeling treatment of the protected pool is first-order, and it is exactly where the policy ranking demands caution.

(ii) Endogenous SME entry attenuates a 50–60 percent larger latent shock. Hold the SME pool at its pre-policy size while imposing the SME-only rule and the simulated price effect rises to 152 percent of the realized one in non-pharma and 157 percent in pharma. Without endogenous entry the set-aside would cost 50–60 percent more in price and welfare. Participation is the government's partial offset, not the source

of the markup: the SMEs the protected segment induces to enter absorb part of the within-auction shock, though not all of it.

(iii) The price effect operates predominantly within the auction. Let p_{S_1} be the simulated winning price under the pre-policy open regime, p_{S_2} the SME-only counterfactual at the pre-policy SME pool, and p_{S_3} the SME-only counterfactual at the observed post-policy SME pool. In the main specification, $p_{S_3} - p_{S_1} = +0.259$ of the reference price in non-pharma and +0.308 in pharma (Table D.21). The within-auction piece—which I call *sheltered bidding*—is 74.5 percent of that effect in non-pharma and 73.3 percent in pharma. The share stays in the 73–85 percent range across cost-distribution estimators (Turnbull NPMLE: 74, 82) and across the two specifications of the post-policy SME pool (strict invariance: 85, 79). The set-aside's price cost is mostly a property of how the surviving pool bids, not of how many firms it contains.

Three contributions follow. First, the paper provides a within-auction structural decomposition of an SME procurement preference identified from a single legal trigger, implying an annual welfare loss of R$ 55–128 million on a single product group of São Paulo's R$ 13 billion centralized platform. The methodological position rests on a combination of conditions rare in the procurement-preference literature: a total-exclusion legal trigger (clean separation of regimes), a Pregão drop-out auction format (point-identification of cost distributions), an auction-level UH correction (Krasnokutskaya, 2011), an observed-equilibrium-entry treatment in the Athey et al. (2011) spirit, and a Saez and Stantcheva (2016) welfare-weight identity. Marion (2007) estimates the price effect of federal SME set-asides via reduced-form variation; Krasnokutskaya (2011) and Athey et al. (2011) decompose California highway and timber bid preferences into bidding and entry channels via cross-equation constraints. This paper extends the second approach to a total-exclusion preference rule on a single clean legal trigger, allowing the welfare ranking against alternative preference instruments to be computed jointly within

the same framework.

Second, the paper shows that endogenous SME entry is not a side issue but part of the welfare object: a fixed-pool counterfactual—the standard reduced-form shortcut—misstates the realized policy cost by roughly half. Third, it converts the welfare comparison into a market-design statement rather than a universal ranking. The 10 percent price preference welfare-dominates the set-aside in thick standardized markets across both specifications; in thin heterogeneous markets the comparison depends on how entry selection is modeled. The bifurcation identifies precisely the market conditions under which equilibrium-selection vs. primitive-invariance treatment of the protected pool is first-order, and converts a sensitivity check into a substantive claim about where bidder exclusion delivers proportional returns and where it does not.

The paper sits in dialogue with three lines of empirical work: recent procurement-design papers on emerging-market public contracting (Decarolis et al., 2025; Coviello et al., 2026; Ferraz et al., 2016; Szerman, 2023), the political economy of public contracting (Colonnelli and Prem, 2022), and the public-finance evaluation of government programs in developing economies (Best et al., 2023; Bandiera et al., 2021; Olken, 2007). The contribution here is to separate the within-auction and participation channels of an SME procurement preference on a single clean legal trigger, where the partial-preference structural literature has previously relied on cross-equation constraints to disentangle them.

A pair of comparable furosemide 40 mg purchases by the same São Paulo public buyer eight months apart and bracketing the cutoff fixes the magnitudes. In February 2018 the unit cleared roughly twelve percent below the buyer's reference price; in October 2018 it cleared just above. The difference is one realization of the average direction the design estimates across the 297,967 firm-auction observations and 97,993 distinct auctions of the 18-month structural sample on each side of the March 2018 cutoff—10–11 percent on the

absolute final price in the difference-in-differences benchmark of Appendix Appendix I, the magnitude the structural exercise of this paper interprets and prices. The headline numbers across the four reporting layers are summarized in Table 1.

Table 1: Headline magnitudes.

	Non-pharma	**Pharma**
DiD on $\log p^{\text{final}}$ (absolute, App. Appendix I)	+0.10 to +0.11	
DiD on $p^{\text{final}}/p^{\text{ref}}$ (App. Appendix I)	+0.06	
Simulated $p_{S_3} - p_{S_1}$, units of p^{ref}	+0.259	+0.308
Within-auction share, main specification	74.5%	73.3%
Within-auction share, Turnbull NPMLE	74.0%	82.0%
Within-auction share, strict primitive invariance	85.0%	79.0%
Welfare cost (% of p_{S_1}, $\lambda = 0.30$)	28.7%	47.0%
$w_\star^{\text{SME}}$ to prefer set-aside, main spec.	2.42	2.61
$w_\star^{\text{SME}}$, strict invariance	> 1	0.7

DiD magnitudes from the difference-in-differences benchmark in Appendix Appendix I (against 76 never-treated product groups, matched window of 18 months on each side of the March 2018 cutoff). Structural magnitudes from the asymmetric IPV counterfactual under observed equilibrium entry; sources by row: Sec. 6 (decomposition), Sec. 8 (Turnbull NPMLE; strict invariance), Sec. 7 (welfare cost), Sec. 9 (welfare weights). The pharmaceutical welfare-dominance ranking against a 10 percent price preference is robust under the main specification but reverses under strict primitive invariance ($w_\star^{\text{SME}} = 0.7 < 1$); the non-pharmaceutical ranking is preserved under both specifications.

2. Institutional Background

The reform that anchors this paper has a simple top-down structure. A generic legal opinion at the São Paulo state Attorney General's office (PGE-SP) reinterpreted a federal procurement-rule threshold in late 2017; the state's centralized procurement platform (BEC-SP) operationalized the new interpretation across all product groups; and the state audit court (TCE-SP) ratified it five months after the empirical cutoff of this paper. The rest of this section explains why the interaction of this transversal reinterpretation with the typical structure of medical-supplies procurement notices mechanically unlocked Group 65 from the SME-only default, providing the natural experiment exploited below. Table 2 reconstructs the chronology.

Brazil's federal SME statute (Complementary Law 123/2006, amended for procurement by LC 147/2014 and for revenue brackets by LC 155/2016) defines micro and small enterprises by annual gross revenue ceilings of R$360,000 (US$103,000) and R$4,800,000 (US$1.37 million), respectively, and mandates exclusive SME tenders for items valued at R$80,000 or less. LC 147/2014 converted the SME-only rule from optional ("may") to mandatory ("must"), with departures requiring a costly justification to the audit courts; state-level adherence to SME-only tendering for eligible items rose from roughly 13% to nearly 70% in the years that followed.[1]

The state of São Paulo has operated the Bolsa Eletrônica de Compras (BEC) since September 2000 (Decree 45.085/2000), and the Pregão electronic reverse-auction format that anchors this paper's analytical sample was regulated within BEC by Decree 49.722/2005. BEC centralizes Pregão for standardized-goods procurement by state bureaus and affiliated units; the structural sample covers 4.8 million item-level transactions across all product groups over the 36-month estimation window, of which Group 65 accounts for roughly 27 percent. Its catalog is organized in three tiers—group, class, and item—so that "medical, dental, and hospital equipment and supplies" form Group 65, within which class 6531 identifies medications subject to price ceilings set by the Câmara de Regulação do Mercado de Medicamentos (CMED).[2]

Group 65 was a singular exception to the 2014 rule, but for a transversal legal reason rather than a sector-specific one. Until late 2017, the prevailing interpretation of art. 48, I

[1]Adherence figures and Group-65 sub-rates are author tallies from the BEC microdata extract used in the structural sample (Section 3; replication numbers in Table B.8), restricted to items satisfying the federal R$80,000 threshold. The R$4.8M EPP ceiling raised by LC 155/2016 has no direct bite on the structural sample, where SME status is identified from the BEC *enquadramento* flag and the Receita Federal *porte* register at the firm-auction level; the bracket change is reported here for institutional completeness.

[2]Brazilian institutional terms used throughout the paper (Pregão, Convite, BEC, CADMAT, CMED, PGE-SP, TCE-SP, Sefaz-SP) are defined in Appendix Appendix A. Platform institutional description at `www.bec.sp.gov.br/BECSP/Quem_Somos/Quem_Somos_BEC_novo.aspx` (retrieved April 2026); the 4.8-million item-level count and the 860,000 purchase-offer count are author tallies from the same BEC extract used in the structural analysis.

of LC 123/2006 was that the R$80,000 threshold for SME-only tenders applied to the *total estimated value of the procurement notice*, not to each individual item or lot—a reading sustained by TCE-SP in eTC-5509.989.15 (23 September 2015, by tie-breaking vote). Group-65 procurement notices systematically combine small individual items (medications, supplies, dressings) with large notice totals reflecting bulk public-sector purchasing, so the totalizing reading mechanically barred Group 65 from the SME-only default—not because medical supplies were classified as "strategic," but because their typical procurement structure was exactly what the threshold-on-total rule excluded.

On 11 December 2017, PGE-SP issued **Parecer Sub-G Cons. nº 151/2017**, holding that the R$80,000 threshold applies item-by-item—allowing mixed-exclusivity procurement notices in which items below R$80,000 are SME-only and items above remain open—and that the more generous federal regime introduced by LC 147/2014 supersedes the previously applicable São Paulo state acts (Lei Estadual 13.122/2008 and Decree 54.229/2009) that had embedded the restrictive interpretation in state-level practice.[34]

Operationally, BEC moved before PGE consolidated the legal interpretation. COMUNICADO BEC nº 02/2017 (18 July 2017) introduced the SME-only OC functionality, and COMUNICADO BEC nº 03/2017 (20 November 2017) expanded it to mixed-exclusivity Pregão notices combining ample-participation, 25%-quota, and SME-only items in the same notice—the precise operational mechanism that Parecer 151/2017 ratified three

[3]Parecer Sub-G Cons. nº 151/2017, Subprocuradoria-Geral de Consultoria da PGE-SP, ementa: *"Licitações exclusivas para ME e EPP (artigo 48, I, LC nº 123/2006) devem ser realizadas para itens ou lotes cujo valor estimado, individualmente, não ultrapasse o limite legal de R$80.000,00,* ***independentemente do valor total da Oferta de Compra****. . . . Aplicabilidade, na íntegra, da normativa federal."*

[4]Disclosed at the BEC/SP legislation portal (https://www.bec.sp.gov.br/becsp/Legislacao/UI_Selecao.aspx, *Pareceres* section, retrieved April 2026); the Boletim CEPGE indexes only Pareceres da Procuradoria Administrativa, so the BEC portal is the canonical channel for Sub-G Cons. pareceres.

weeks later.[5] The empirical cutoff of March 2018 therefore captures the moment of mass take-up of the new functionality by Group-65 PBUs rather than formal regulatory enablement (which was incremental between July and November 2017); SME-only adherence within Group 65 jumped to 43% on average in the months following the cutoff, below the rate for other groups largely because class-6531 pharmaceuticals are oligopolistic and often fail the three-eligible-SMEs requirement.

What matters for identification is that the trigger was a generic statutory reinterpretation, and the unlocking of Group-65 set-asides was a mechanical consequence of how medical-supplies notices are structured. PGE-SP's parecer made no substantive claim about medical supplies or any other product group; it pronounced on a general legal question, with timing set by the bureaucratic process inside PGE-SP and by when BEC operationalized the new interpretation. The TCE-SP reversal in May 2018 lies after the cutoff and confirms that the trigger was a top-down legal interpretation: when the audit court itself flipped, it ratified what PGE-SP and BEC had already established for state-level procurement rather than initiating an independent change in the Group-65 environment. The change is therefore exogenous to Group-65 market conditions on the dimensions the design tests. This is the institutional basis for Assumption 3: the cutoff shifts admissibility and equilibrium entry selection, not the underlying production technology of the goods themselves. Three empirical checks corroborate the institutional reading: the reduced-form DiD against the 76 never-treated product groups (Appendix Appendix I); the primitive-invariance KS test on non-SME F_c (Table C.12); and the cross-modality consistency between Convite GPV inversion and Pregão drop-out recovery in the unaffected pharma non-SME Pre stratum (Figure 1).

[5]Both Comunicados BEC are disclosed at the BEC/SP legislation portal cited above (*Comunicados* section). TCE-SP itself reversed its restrictive reading on 16 May 2018 in eTC-9589.989.18 (Tribunal Pleno), aligning with PGE-SP's expansive interpretation two and a half months after the empirical cutoff of this paper. Both TCE-SP rulings are accessible at the court's jurisprudence portal (https://www.tce.sp.gov.br/jurisprudencia).

Table 2: Institutional timeline—federal and São Paulo state acts relevant to the structural sample.

Date	Act	Content
Sep 2000	Decree 45.085/2000 (SP)	BEC platform created; renamed Decree 45.695/2001.
Jun 2005	Decree 49.722/2005 (SP)	Pregão electronic reverse-auction regulated within BEC.
Dec 2006	LC 123/2006 (federal)	SME statute (*Estatuto Nacional*).
2008	Lei Estadual 13.122/2008 (SP)	SP state-level SME procurement regulation (less favorable than federal post-2014).
2009	Decree 54.229/2009 (SP)	Regulates Lei Estadual 13.122/2008.
Aug 2014	LC 147/2014 (federal)	SME-only default ("may"→"must"); R$80k threshold.
Sep 2015	TCE-SP eTC-5509.989.15	Restrictive reading of art. 48, I (R$80k on notice total) sustained by tie-breaking vote.
Oct 2016	LC 155/2016 (federal)	EPP revenue ceiling R$3.6M→R$4.8M (effective Jan 2018).
18 Jul 2017	COMUNICADO BEC nº 02/2017	SME-only OC functionality enabled in BEC.
20 Nov 2017	COMUNICADO BEC nº 03/2017	Mixed-exclusivity (Ample / 25% quota / SME-only) item-by-item enabled.
11 Dec 2017	PGE-SP Parecer Sub-G Cons. 151/2017	Item-by-item R$80k reading; federal regime supersedes Lei 13.122/2008.
Mar 2018	Mass adoption (empirical cutoff)[a]	Group-65 PBU take-up of the new functionality (first wave of adopting PBUs).
16 May 2018	TCE-SP eTC-9589.989.18	Tribunal Pleno reverses, aligns with PGE-SP expansive reading.
Apr 2021	Lei 14.133/2021 (federal)	New procurement statute; SME defaults preserved.

[a] BEC formally enabled the new functionality on 18 July 2017 (Comunicado 02/2017) and expanded it to item-by-item mixed exclusivity on 20 November 2017 (Comunicado 03/2017). The empirical cutoff in March 2018 (BEC purchase-order sequence index data_oc_numb = 698) corresponds to the moment of mass take-up by Group-65 PBUs, not formal enablement; the gap reflects the diffusion lag between operational availability and PBU adoption. The TCE-SP reversal in May 2018 (eTC-9589.989.18) post-dates the cutoff, confirming that the trigger of the design is the PGE-SP/BEC operationalization sequence, not the audit-court realignment.

3. Data and Sample Construction

The data come from BEC, the state of São Paulo's centralized electronic procurement platform. BEC intermediates all standardized-goods procurement for state bureaus and affiliated units, so the raw extract covers the universe of item-level bids, transacted prices, firm identifiers, quantities, and auction identifiers for January 2016 through December 2019—3.7 million bid-level observations across 82,569 items, 832,984 purchase orders, and 1,344 public buyer units (a 36-month subset of the 4.8 million item-level transactions cited as the platform throughput in Section 2). Group 65 (medical, dental, and hospital supplies) accounts for roughly 27 percent of transactions over the window and is the treated group; the remaining 76 product groups form the never-treated control set in the difference-in-differences benchmark of Appendix Appendix I.

The structural analysis narrows to a sharper sample. Identification uses a symmetric window of 18 months on each side of the March 2018 cutoff (36 months in total, the "18m" label of Table E.30); in Stata monthly-date units the window is $[680, 715]$, or September 2016 through August 2019 inclusive. Within that window, I restrict to Pregão auctions in Group 65. The restriction does two things. Pregão is the descending-clock format that admits point identification of cost distributions from drop-out bids, so the structural estimation requires it; Group 65, in turn, is where the policy bound—the only product group to switch regime during the sample.

Bids are organized at the firm-auction level. Each observation is the final bid submitted by one firm in one item-auction, together with a winner/loser flag. I apply three filters. First, bids must satisfy $c_\epsilon = b/p^{\text{ref}} \in (0, 3]$; the upper cutoff removes implausible outliers (less than 0.4 percent of bids, almost always data-entry errors at the hundred-times-reference level). Second, auctions must have at least two participating firms—below that, neither point identification of F_c nor the BNE counterfactual is well-defined. Third, each bid carries a type flag (SME or non-SME, from the BEC administrative SME

registry, validated against historical bidding to avoid misclassifying firms that temporarily exited SME status) and a pharma flag (CADMAT classes 6531, 6532, 6536, 6581—CMED-regulated pharmaceuticals and biologicals—are pharma-narrow; the remainder of Group 65 is non-pharma).

The resulting structural sample is 297,967 firm-auction observations across 97,993 distinct auctions, balanced across Pre (September 2016 to February 2018, 18 months, $n = 48{,}740$ auctions) and Post (March 2018 to August 2019, 18 months, $n = 49{,}253$). Descriptive counts by stratum appear in Appendix Tables B.9 and B.10.

Auction-level unobserved heterogeneity is material and not fully absorbed by the reference-price normalization. Two items inside the same CADMAT class can share a scale component that moves every bid in the auction together—volume differences, ANVISA registration burden, contract-specific complexity. I handle this with the Krasnokutskaya (2011) method-of-moments variance decomposition followed by best-linear-predictor shrinkage; details in Section 5, with the recovered intraclass correlations and the variance panel in Appendix Table C.14. The cleaned residuals $c_\epsilon^{\text{clean}} = \exp(\hat{e}_{it})$ enter all downstream structural estimation.

Entry counts enter separately. For each auction I count the distinct firms of each type that submitted at least one bid, then average within (period $\times$ pharma) cells. Entry rates enter the BNE as observed equilibrium arrivals (Section 5); the panel is Appendix Table D.19, with magnitudes interpreted in Section 6.

The analysis relies on four ancillary inputs. Reference prices come from the BEC catalog (`preco_ref`) and are used both for normalization and for pricing out the fiscal cost of the rule in reais. The reference price for each item is set ex ante by the buyer from the BEC catalog and does not move with the regime change: item-level fixed effects in the DiD specifications and the structural normalization absorb cross-sectional variation in p^{ref} across items, and the BEC catalog architecture sets p^{ref} at the item level for the

entire window. The variable that changes around the cutoff is regime admissibility, not item-level catalog pricing. CADMAT class codes come from the BEC product hierarchy. The historical SME flag is reconstructed from firm-level bidding histories over the full 2014–2019 period to protect against transient status changes. The pharma flag follows the CADMAT class list above.

The pair of furosemide 40 mg purchases mentioned in the introduction (item 110639, a CMED-regulated generic antihypertensive) sits in the institutional sample. February 2018: ten firms bid, clearing price 12 percent below reference. October 2018: three SMEs bid, clearing price just above reference. Welfare arithmetic on this pair appears in Section 7; the structural averages are over 59,997 Pregão pharmaceutical auctions in the 18-month sample (Table B.9).

4. Model

A stylized model captures what the set-aside does and, as important, what it does not do. N potential bidders compete for a single indivisible contract. Each bidder i has type $k_i \in \{\text{SME}, \neg\}$ and draws a private cost of supply c_i from a type-specific distribution F_c^k, with F_c^{SME} and $F_c^{\neg}$ differing in ways the data are allowed to determine.

The auction format is the identifying payload. BEC implements procurement through a Pregão—a descending-clock electronic English-reverse auction, distinct both from first-price sealed-bid designs (Convite, in the smaller-value Brazilian segment) and from the Dutch descending-price format more commonly studied in the auction literature. The clock opens at the buyer's reference price p^{ref}, firms post downward-revisable offers until only one firm remains, and that firm supplies the contract at the clearing price. What makes this format analytically convenient is a standard result of Haile and Tamer (2003): under independent private values, the weakly dominant strategy is to exit the auction when the running clock price reaches one's own cost. Drop-out prices of losers therefore

point-identify their costs, without requiring the researcher to invert a bid function or assume a parametric form for the bidding strategy. The winner, who never drops out before the clock stops, leaves only an upper bound $c_{\text{win}} \leq c_{(2)}$ (the cost of the second-to-last firm to exit)—and that upper bound is exactly what the Turnbull robustness check in Section 8 uses to deliver a non-parametric MLE of F_c^k. This point-identification under Pregão is the reason a structural analysis is available here where the cross-sectional first-price-sealed-bid settings of Marion (2007), Krasnokutskaya (2011), and Athey et al. (2011) have relied on cross-equation constraints to disentangle channels.

The model relies on three substantive restrictions, stated formally below and defended in turn with the diagnostic evidence in Section 5.

Assumption 1 (Independent Private Values, asymmetric across types). *Costs c_i are drawn independently across bidders. Conditional on type k, $c_i \sim F_c^k$ with density f_c^k on support $[\underline{c}, \overline{c}] \subset \mathbb{R}_+$. The SME and non-SME distributions are not constrained to coincide.*

Assumption 1 permits SMEs and non-SMEs to differ in cost distributions—a realistic feature given that non-SMEs are typically the large distributors of the same generic pharmaceutical—without requiring one to stochastically dominate the other. The empirical median-cost gap of 0.10–0.30 of the reference price across (pharma $\times$ type) cells documented in Section 5 supports asymmetry directly; symmetric IPV would require this gap to vanish, which the data reject.

Assumption 2 (Multiplicative auction-level heterogeneity). *The normalized log-bid decomposes as $y_{it} = \log(b_{it}/p_t^{\text{ref}}) = a_t + e_{it}$, where a_t is an auction-level scale common to all bidders in auction t, and e_{it} is bidder-specific and independent across bidders within t. Both a_t and e_{it} are mean-zero with finite variances $\sigma_a^{2,k}$ and $\sigma_e^{2,k}$ that may vary by stratum.*

Assumption 2 is the Krasnokutskaya (2011) restriction that separates a common scale from the bidder-specific private value. Multiplicativity is the natural form when the auction-level shock affects all bidders proportionally to their cost realizations—standard

in procurement settings where item complexity, ANVISA registration burden, or contract specifications scale costs uniformly across firms. It is weaker than the fully non-parametric Kotlarski (1967) deconvolution, which would additionally require type-symmetry within e_{it}; Section 8 reports a Turnbull NPMLE that dispenses with Assumption 2 altogether and corroborates the baseline numbers within 16 percent of the total effect across classes (with the largest gap in pharmaceutical Turnbull).

Assumption 3 (Set-aside, with equilibrium SME selection). *Under the open regime both SMEs and non-SMEs enter with strictly positive probability. Under the SME-only regime, only SMEs are admissible. Within a short window around the cutoff, the underlying technology of supply is stable, but the equilibrium distribution of active SMEs under the restricted regime may differ from its pre-regime counterpart because the policy changes which SMEs find entry profitable. Write these distributions $F_c^{SME,\tau}$ for $\tau \in \{Pre, Post\}$.*

Proposition 1 (Identification of F_c^k under ascending-clock IPV). *Under Assumptions 1–3, the type-specific cost distribution F_c^k is point-identified from drop-out exit prices in the Pregão English-reverse format under Haile and Tamer (2003), up to the auction-level scale a_t recovered by the Krasnokutskaya (2011) variance decomposition of Assumption 2. The post-policy SME cost distribution $F_c^{SME,Post}$ is identified as an equilibrium-selection object under Assumption 3, distinct from but nested in the strict-invariance limit $F_c^{SME,Post} = F_c^{SME,Pre}$.*

Assumption 3 is the structural counterpart of the parallel-trends assumption in reduced-form DiD, but with equilibrium selection made explicit. In the Athey et al. (2011) reading adopted here, the policy changes the admissible set and therefore the composition of the SME bidders who actually enter. The post-policy SME cost distribution is thus an equilibrium object, not evidence of a primitive technological break. This interpretation fits the diagnostic evidence: the primitive-invariance KS test on non-SME F_c (Table C.12) rejects strict invariance in the pharma Pregão stratum ($D = 0.0722$) and in the non-pharma Convite stratum ($D = 0.0514$), and fails to reject in the pharma Convite ($D = 0.0324$) and non-pharma Pregão ($D = 0.0486$) strata. The operational threshold $D < 0.05$ corresponds to less than five percentage points of CDF movement at the

Kolmogorov sup norm. The cross-modality check between Convite GPV and Pregão drop-out recoveries in Table C.13 provides an additional consistency test.

Strict invariance as a limiting case. A stronger reading would impose $F_c^{\text{SME,Post}} = F_c^{\text{SME,Pre}}$ and attribute the entire policy response to bidder counts alone. I treat that as a useful benchmark rather than the main specification. Section 8 reports it explicitly as a strict-primitive-invariance limit: it preserves the intensive-margin headline, but it changes the welfare comparison in the thin, heterogeneous pharma market because it shuts down equilibrium selection in who enters.

What the KS rejection of strict invariance implies for the modeling choice. The KS tests in Tables C.12 and C.16 reject strict invariance of non-SME F_c in three of the four Pregão strata after UH cleaning, with KS distances of 0.0613 (non-pharma Pregão) and 0.1407 (pharma Pregão). This rejection is not a problem for the structural exercise; it is the empirical foundation for adopting equilibrium selection (Assumption 3) as the main specification rather than strict invariance. A model that imposed strict invariance and then ran the BNE counterfactual on data that visibly violate it would be misspecified by construction. The main-specification approach taken here—to treat $F_c^{\text{SME,Post}}$ as an equilibrium object generated by the protected regime and to compare results against the strict-invariance bound in robustness—is exactly the response the KS evidence calls for. The rejection of strict invariance is therefore evidence in favor of the modeling choice, not against it.

Equilibrium and counterfactuals. The auction format is interpreted as ascending-clock IPV throughout: under Assumption 1, exit-at-cost is the weakly dominant strategy, the surviving bidder pays the second-lowest exit, and the expected transaction price equals $\mathbb{E}[c_{(2)}]$. Revenue equivalence (Vickrey, 1961; Maskin and Riley, 2000) extends the same expected-price object to FPSB and to other revenue-equivalent formats, but

the operative interpretation in this paper is the ascending clock that BEC's Pregão implements; Krasnokutskaya (2011) provides the analog argument used here. The counterfactuals therefore reduce to Monte Carlo simulation over draws from the estimated F_c^k: draw a stratum-specific count of bidders from each type's Poisson arrival process, draw costs, sort, and read off the two lowest order statistics. The lowest, $c_{(1)}$, is the production cost; the second-lowest, $c_{(2)}$, is the expected transaction price. The exercise is conducted *under observed equilibrium entry*: post-policy participation rates and the post-policy SME cost distribution are taken from data, not derived from a free-entry condition jointly with the cost primitive. The model is therefore not a fully solved nested entry-bidding equilibrium of the kind developed in Krasnokutskaya and Seim (2011), where entry costs are structurally identified from a free-entry indifference condition that jointly disciplines the bidding stage. Here, entry costs are reported as zero-profit calibration margins (Section 6); the decomposition that follows is a counterfactual-simulation accounting under the modeling assumptions stated rather than a separation of independently identified structural channels. The welfare-cost expression of Section 7 prices the simulated price effect at the marginal cost of public funds λ, which enters the model as a planner-side primitive rather than as a quantity recovered from the auction.

Entry. The Bayes-Nash counterfactual takes equilibrium arrival rates as primitives rather than solving for them explicitly. Observed Pre-period counts $(N_{\text{Pre}}^{\text{SME}}, N_{\text{Pre}}^{\neg})$ are the status-quo arrival rates under the open regime; Post-period counts $(N_{\text{Post}}^{\text{SME}}, 0)$ are the equilibrium arrival under the restricted regime. This is a reduced-form treatment of entry in the Athey et al. (2011) spirit; the main specification combines the observed Post counts with the observed Post SME bidder distribution. The implied zero-profit entry costs per bidder, reported separately (Table D.20), are in the R\$0.11–R\$2.46 range. These are per-auction zero-profit margins at the entry margin, not the fixed cost of preparing a BEC documentation packet (which is amortized over many auctions and is

not directly recovered by this calibration); they are a consistency check on the assumed Poisson primitives rather than a structural estimate.[6]

The decomposition. Three counterfactual scenarios anchor the exercise:

- S_1: open regime at the pre-period pool. The baseline.
- S_2: SME-only at the pre-period SME pool. The non-SME type is removed; the SME pool is held at its observed pre-policy arrival rate. Within-auction component isolated.
- S_3: SME-only at the post-period SME pool. Uses the observed post-policy arrival rate and the post-policy SME cost distribution. Full policy effect with endogenous entry.

The simulated total effect on the second-order statistic decomposes arithmetically as

$$\underbrace{p_{S_3} - p_{S_1}}_{\text{simulated total effect}} = \underbrace{(p_{S_2} - p_{S_1})}_{\text{within-auction component (sheltered bidding)}} + \underbrace{(p_{S_3} - p_{S_2})}_{\text{participation-margin component}} . \quad (1)$$

Both components are properties of the simulation given F_c^k and the observed entry pattern; they sum to the simulated total effect by construction. The within-auction component can exceed the realized total effect when the participation-margin component partially offsets it, which is what the data deliver in non-pharmaceutical markets.

Why "sheltered bidding". I refer to the within-auction component as *sheltered bidding*: a descriptive label for the change in the simulated $\mathbb{E}[c_{(2)}]$ when non-SMEs are removed and the SME pool is held at its pre-policy size. The label captures the equilibrium

[6]Independence-across-auctions is assumed at the level of cost draws conditional on type and stratum. Within-auction correlation enters the model through the auction-level scale a_t (Assumption 2); the cluster bootstrap of Section 6 resamples at the auction level to preserve this correlation in the standard errors.

price formation under reduced competition, not any active behavioral adjustment by SMEs—under ascending-clock IPV, the dominant strategy is exit-at-cost regardless of pool composition, and the within-auction component is mechanically a property of which order statistic is realized at each pool composition. The share of the simulated total effect attributable to the within-auction component depends on the cost-distribution estimator and on whether one imposes strict invariance on the post-policy SME pool; reading the share as a tightly pinned point estimate would be misleading. The pharmaceutical decomposition is reported alongside as a more model-sensitive boundary case.

Scope of the structural exercise. The model deliberately stops short of five extensions. (i) Entry is taken as observed Poisson arrival rates, not solved jointly with bidding under a free-entry condition. (ii) The implied κ^k are zero-profit margins at the entry margin, not structurally identified fixed costs. (iii) Procurement quantity is treated as inelastic at observed values, with the implied bound on the welfare cost discussed in Section 9. (iv) The 10 percent preference V3 is computed under Vickrey-equivalent translation; the FPSB equivalent under Maskin and Riley (2000) adjustment is bounded above by the type-specific cost-distribution dispersion in each stratum, with the welfare ranking V3 $\succ$ V0 preserved at the upper bound (Section 8.9). (v) Asymmetric IPV is assumed; symmetric IPV is consistent with the data only at very low quantile dispersion across types and is not formally tested. Each is a margin a fully solved structural model would price; each is bracketed below or in robustness rather than imposed as the central exercise.

5. Identification and Estimation

Three objects require estimation: the type-specific cost distributions F_c^k, the auction-level scale a_t, and the equilibrium entry counts. The third is directly observed. The first

two require structure—the first by design of the Pregão auction, the second through the variance decomposition in Assumption 2.

5.1. Type-specific cost distributions

Under Assumption 1 and the English-reverse interpretation, the drop-out price of each loser is a point observation of that loser's cost. The empirical distribution of the normalized drop-outs within a stratum would suffice as an unbiased estimator of F_c^k if the winner could be observed; absent that, losers-only omits the lowest cost draw per auction and biases F_c^k systematically upward at the left tail. I therefore use an *all-bidders* ECDF, which adds the winner's final clock price to the stratum-specific ECDF. In a descending-clock auction this final price is approximately $c_{(2)}$, the second-to-last exit—an upper bound on the winner's cost but a substantially tighter one than the zero imputed by losers-only omission. The bias in all-bidders runs in the opposite direction and is materially smaller: losers-only overstates F_c^k in the left tail; all-bidders understates it slightly throughout. The Turnbull NPMLE of Section 8 eliminates both biases by treating the winner as left-censored at $c_{(2)}$, delivering a strict ordering $F_c^{\text{Turnbull}} \leq F_c^{\text{all-bidders}} \leq F_c^{\text{losers-only}}$ in the left tail; the three coincide asymptotically.

A Turnbull NPMLE, reported in Section 8, treats the winner formally as left-censored at $c_{(2)}$ and delivers point-identified F_c^k without either bias. Across the eight (class $\times$ type $\times$ period) strata (Table E.24), the median estimates satisfy Turnbull $\leq$ all-bidders $\leq$ losers-only in all 8 strata, and the 75th-percentile estimates satisfy all-bidders $\leq$ Turnbull $\leq$ losers-only in 6 of 8 strata (the two pharmaceutical non-SME cells reverse, with Turnbull marginally below all-bidders by 0.01–0.02 of the reference price; this is within the bootstrap CI and does not reverse the substantive ordering of estimators in either cell). The BNE decomposition computed on the Turnbull input differs from the all-bidders baseline by at most 16 percent of the total effect across classes (Table E.25), with the largest gap in pharmaceutical Turnbull. All-bidders is the main-text estimator;

Turnbull is the robustness benchmark; losers-only is the input the DiD benchmark of Appendix Appendix I effectively absorbs through the item fixed effects.

A cross-modality consistency check supports the descending-clock specification on a single testable hypothesis: under correct asymmetric-IPV specification, the Guerre et al. (2000) inversion of Convite first-price bids and the Haile and Tamer (2003) drop-out point-ID of Pregão exits should recover the same F_c^k in the stratum unaffected by the policy reform. The data deliver convergence to within one percentage point at the median in pharma non-SME Pre—the stratum that predates the cutoff and is staffed by non-SMEs unaffected by the set-aside. In other strata the gap is wider and motivates the auction-level heterogeneity correction below. Figure 1 plots the two recoveries side by side for all four strata. The cross-modality convergence is consistent with non-collusive bidding in the pharma non-SME Pre stratum—two independent structural recoveries of the same primitive land within sampling error of each other, ruling out systematic divergence between FPSB and English-reverse formats of the kind Conley and Decarolis (2016) bidder-group detection tests would flag. Complementary bid-pairing screens reported in Section 8.8 probe within-format coordination patterns at the firm-pair level; the cross-modality and pair-pattern checks are not contradictory. A formal test of $F_c^{\mathrm{SME}} = F_c^{\neg}$ (symmetric IPV against the asymmetric reading) is not the central exercise, but the recovered cell-level medians and 75th-percentile estimates differ by 0.10–0.30 of the reference price across the (pharma $\times$ type) cells (Table C.17), well outside Monte Carlo sampling noise; asymmetry is therefore consistent with the data, not just imposed by Assumption 1.

5.2. Auction-level heterogeneity

The median-cost gap across modalities in non-pharma—up to 23 percentage points of the reference price—is too large to attribute to sampling noise. Reference-price normalization absorbs the systematic scale variation across items; the residual within-

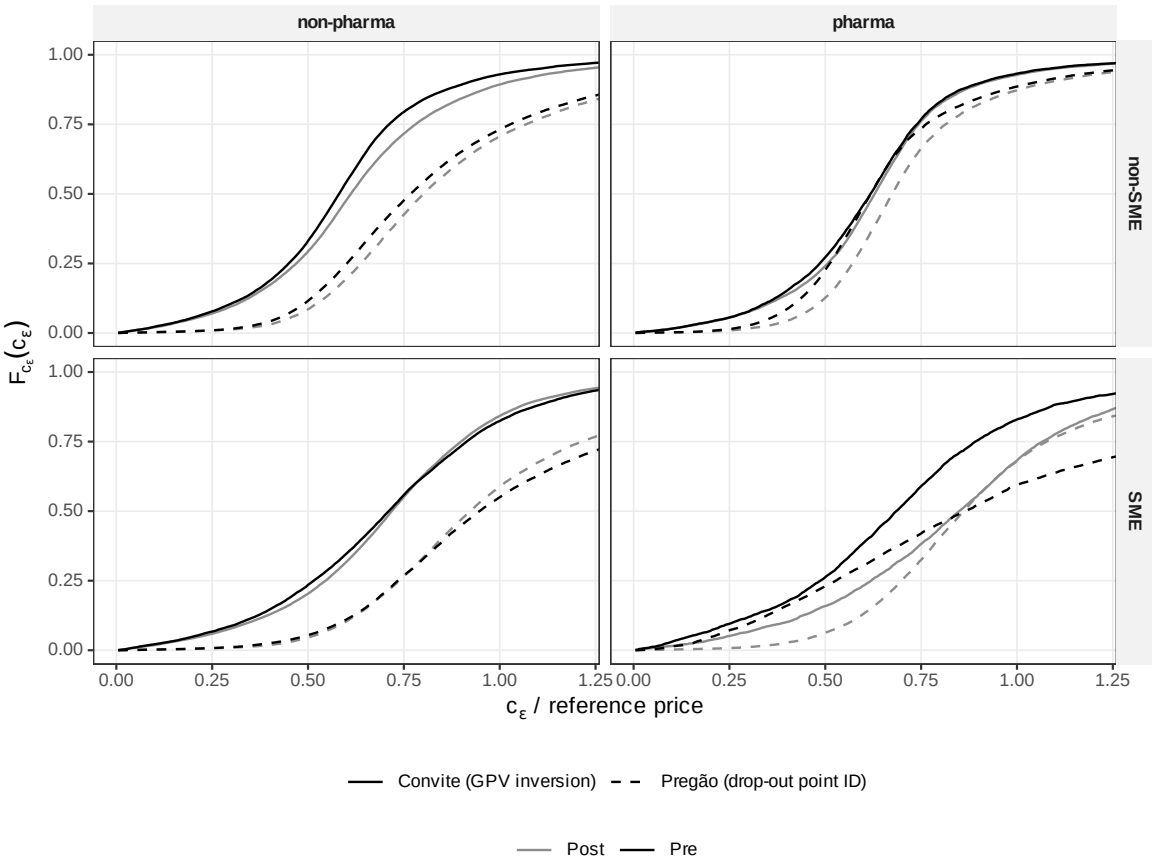

Figure 1: *Cross-modality identification check, UH-clean F_{c_ϵ}.* Each panel plots the type-specific cost distribution estimated two ways on the same sample: the Guerre et al. (2000) inversion on Convite first-price sealed-bid auctions (solid) and the Haile and Tamer (2003) drop-out point-ID on Pregão descending-clock auctions (dashed). Pre and Post periods are colour-coded. Convergence in the pharma non-SME Pre panel (upper right) supports the IPV restriction in that stratum: two independent structural recoveries on the same primitive land within sampling error of each other. Residual gaps in other panels motivate the auction-level heterogeneity correction in Section 5.

class auction-level variance is what Assumption 2's a_t structures. Two items in the same CADMAT class can differ in ANVISA registration burden, in contract complexity, or in volume-specific cost elasticities in ways the reference price does not fully reflect.

I estimate the variance decomposition by method of moments. Within each stratum $(k, \text{pharma}, \text{period})$, $\sigma_a^{2,k}$ is the between-auction variance of auction-mean log-bids, and $\sigma_e^{2,k}$ is the residual within-auction variance. The intraclass correlation

$$\rho^k = \frac{\sigma_a^{2,k}}{\sigma_a^{2,k} + \sigma_e^{2,k}}$$

ranges across the four Pregão (pharma $\times$ type) cells from approximately 0.28 (Pregão pharmaceutical SMEs, average across periods, with a low-n Pre-period stratum pulling the cell mean down) through 0.36 (Pregão non-pharmaceutical SMEs) and 0.39 (Pregão non-pharmaceutical non-SMEs) up to approximately 0.59 (Pregão pharmaceutical non-SMEs); the corresponding Convite cells deliver 0.55 and 0.66 (Table C.14). Restricted to the three Pregão cells outside the low-n pharma-SME-Pre stratum (range 0.36–0.59), the order of magnitude is consistent with the auction-level scale share Krasnokutskaya (2011) reports for highway procurement (a private-information component of approximately 34 percent of total cost variance, implying an ICC of approximately 0.66 in that setting). A best-linear-predictor shrinkage then estimates each auction's specific heterogeneity as

$$\hat{a}_t = \frac{\hat{\sigma}_a^2 \, N_t}{\hat{\sigma}_a^2 \, N_t + \hat{\sigma}_e^2} \cdot (\bar{y}_t - \bar{y}) \,,$$

and the UH-clean residuals $\hat{e}_{it} = y_{it} - \hat{a}_t$ deliver cleaned cost estimates $c_\epsilon^{\text{clean}} = \exp(\hat{e}_{it})$. These cleaned bids are the inputs to all downstream structural estimation.

The correction closes ninety-three percent of the cross-modality median gap in pharma SME Post and twenty-three percent in pharma non-SME Post; the residual non-pharma gap (approximately 17 percentage points after UH cleaning) is what the sample filter

$c_\epsilon \leq 3$ partially absorbs. Auction-level heterogeneity is therefore the binding constraint in pharma but only part of the story in non-pharma, where richer item-level heterogeneity presumably operates. The Krasnokutskaya shrinkage assumes Gaussianity of a_t and e_{it}; the Turnbull NPMLE in Section 8 dispenses with this assumption and ratifies the estimates.

5.3. Entry

Entry is modeled as Poisson arrivals with stratum-specific rates. I take the observed Pre-period average counts of SME and non-SME bidders per auction as the status-quo arrival rates, and the Post-period SME counts as the equilibrium arrival under the restricted regime. No structural entry cost is jointly estimated with the cost primitives; this is the reduced-form treatment in the Athey et al. (2011) spirit. The same logic applies to the post-policy SME bidder distribution: in the main specification, $F_c^{\text{SME,Post}}$ is treated as an equilibrium-selection object generated by the protected regime rather than forced to equal $F_c^{\text{SME,Pre}}$. Section 8 reports the stricter limiting case in which that equality is imposed as a benchmark.

Implied entry-cost magnitudes. A zero-profit calculation applied to the simulated winning-margin profit distribution delivers an implied entry cost per bidder in the R$0.11–R$2.46 range (Table D.20). Non-SMEs face roughly 4.5 times the SME entry cost across both classes ($\kappa^{\neg}/\kappa^{\text{SME}} = 4.5$ in non-pharmaceuticals and 4.6 in pharmaceuticals). The asymmetry is consistent with the documentation and certification burden of responding to a BEC tender, materially heavier for larger firms with broader product portfolios subject to ANVISA regulatory scrutiny; the per-auction marginal opportunity-cost calibration that anchors these magnitudes is reported in the Section 6 entry-cost footnote.

5.4. Counterfactual simulation

With F_c^k and entry counts in hand, the BNE counterfactual is a Monte Carlo over draws from the estimated primitives. For each of $B = 2{,}000$ simulated auctions, I draw $n^{\text{SME}} \sim \text{Poisson}(\bar{N}^{\text{SME}})$ and analogously for non-SMEs (the Poisson rate is the observed mean entry count by stratum, distinct from the MCPF λ used in the welfare arithmetic of Section 7), sample costs from the corresponding F_c^k, sort, and record $c_{(1)}$ (the winner's cost, the production cost), $c_{(2)}$ (the expected transaction price under revenue equivalence), and the winner's type. Aggregating over draws yields the means, quantiles, and decompositions reported in Section 6. Standard errors come from a cluster bootstrap at the auction level with $B_{bs} = 500$ replicates; details are in Section 8.

6. Results

The results organize around three findings. First, the simulated set-aside price effect is mostly within-auction: the within-auction share is 74.5 percent in non-pharma and 73.3 percent in pharma, and stays in the 73–85 percent range across cost-distribution estimators.[7] Second, endogenous SME entry is a partial offset, not the source of the markup: the induced participation response attenuates the within-auction shock, and without it the realized price and welfare cost would be 50–60 percent larger. Third, the policy comparison between bidder exclusion and a 10 percent price preference is conditional on goods characteristics: in thick standardized non-pharma markets the preference rule welfare-dominates robustly; in thin heterogeneous pharma the ranking depends on how the post-policy entrant mix is modeled.

[7] The DiD-vs-structural relationship is reconciled in Section 6.4 below.

6.1. Entry response

Table D.19 reports the panel. Pre-policy Pregão has 0.94 SMEs and 2.68 non-SMEs per auction in non-pharma, 0.55 and 2.61 in pharma. Post-policy the counts rotate: 1.87 SMEs and 1.50 non-SMEs in non-pharma, 1.22 and 1.66 in pharma. SMEs roughly double in both classes. Non-SMEs fall sharply in both classes. The set-aside therefore changes both the admissible pool and the equilibrium participation response.

6.2. Structural decomposition of the set-aside

Table D.21 gives the point estimates under the all-bidders regime and endogenous entry, which is the main specification. In non-pharma, the simulated open-regime price is $p_{S_1} = 0.759$ of the reference price. Holding the SME pool at its pre-period size and restricting the auction to SMEs raises the simulated price to $p_{S_2} = 1.152$, a within-auction shift of $+0.393$. Letting SMEs enter at their post-policy rate pulls the simulated price down to $p_{S_3} = 1.018$, a participation-margin offset of -0.135. The net effect is $p_{S_3} - p_{S_1} = +0.259$.

The pharmaceutical class follows the same decomposition with larger magnitudes. $p_{S_1} = 0.656$, $p_{S_2} = 1.141$, $p_{S_3} = 0.964$, and the total effect is $+0.308$. The within-auction shift is $+0.485$; the participation offset is -0.177. The larger magnitudes line up with the thinner baseline SME pool in pharmaceuticals (0.55 vs 0.94 SMEs per auction Pre-period, Table D.19) and the heavier auction-level heterogeneity (ICC 0.59 vs 0.36 in Pregão non-SME cells, Table C.14).

The within-auction share, defined as $|p_{S_2} - p_{S_1}|/(|p_{S_2} - p_{S_1}| + |p_{S_3} - p_{S_2}|)$, is 74.5 percent in non-pharmaceuticals and 73.3 percent in pharmaceuticals in the main specification. Table E.25 shows 74 and 82 percent under the Turnbull NPMLE. Table E.26 shows 85 and 79 percent under the strict-primitive-invariance benchmark. Across classes and specifications, the within-auction share sits in the 73–85 percent range. The common

message is that the set-aside mainly works by changing the order statistic inside the restricted auction; entry matters, but as a partial offset. Figure 2 makes this decomposition visible.

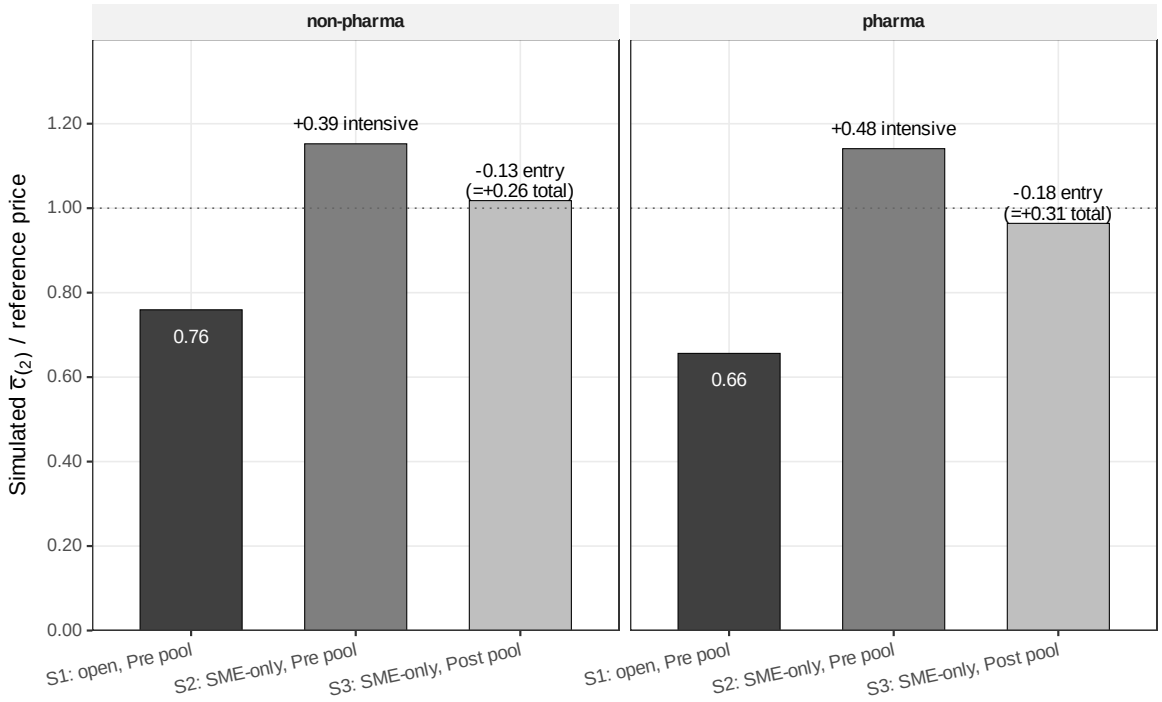

Figure 2: *Bayes-Nash counterfactual price decomposition by class, under observed equilibrium entry.* Bars report the simulated second-order statistic $\bar{c}_{(2)}$ normalized by the item reference price, under three counterfactual scenarios. S_1 is the open regime at the Pre-period pool (baseline). S_2 imposes SME-only at the Pre-period SME pool size (within-auction component isolated). S_3 is the SME-only counterfactual at the observed Post-period SME pool (full policy effect with endogenous entry). Annotations report the within-auction component ($S_2 - S_1$), the participation-margin component ($S_3 - S_2$), and the simulated total effect ($S_3 - S_1$). The dotted horizontal line marks the reference price ($p^{\text{ref}} = 1$). Under the main specification, the within-auction component dominates the simulated total effect in both classes; the pharmaceutical decomposition is more model-sensitive (Section 8). Numbers from Table D.21.

6.3. Endogenous entry as a partial offset

Table F.34 makes the role of entry visible. V2 imposes the full set-aside while holding the SME pool fixed at its pre-policy size. Relative to the realized full-set-aside counterfactual V0, V2 is 152.0 percent of V0 in non-pharmaceuticals and 157.4 percent in pharmaceuticals. The fixed-pool simulated effect therefore exceeds the realized one by roughly 50–60 percent. Both terms in the welfare formula of Section 7 load on this

simulated price margin, so the fixed-pool welfare arithmetic would materially overstate the burden of the policy under the model's main specification.

The substantive content is that the within-auction component of the simulated total effect dominates the participation-margin component in non-pharmaceutical markets, and that the participation response attenuates the resulting burden ex post. Reading the participation response as a partial offset to a larger latent within-auction shock is the substantive claim; reading it as a free-standing fiscal-insurance result would overstate the model. The 50–60 percent attenuation is conditional on the policy instrument: under V3 (10 percent price preference), the analogous fixed-pool-vs-realized comparison delivers an attenuation close to zero, since non-SMEs remain in the auction and the participation margin barely moves. The entry response is therefore a property of the full-set-aside instrument, not a free-standing feature of the SME population. Figure 3 shows the size of the simulated offset.

6.4. Sources of the departure from earlier estimates

Table D.22 reports the 2×2 grid in which each axis is a methodological choice: cost distributions raw versus UH-clean, and entry fixed at the pre-period pool versus endogenous. The clean-and-endogenous row is the main spec; the raw-and-fixed-pool row is the reduced-form-style proxy. The relative gap between them is about 72 percent in non-pharmaceuticals and about 44 percent in pharmaceuticals. UH correction accounts for more than half of the gap by changing the right tail of the recovered cost distribution; endogenous entry accounts for the rest by recognizing the policy-induced thickening of the SME pool.

Table D.23 pushes the same logic further by bridging the DiD coefficient itself to the structural counterfactual. Four mechanical sources of divergence—sample restriction ((a) Pregão only, $c_\epsilon \leq 3$, $n \geq 2$), UH cleaning ((b) Krasnokutskaya scale shrinkage), functional form ((c) second-order statistic vs. linear log-mean), and conditioning set

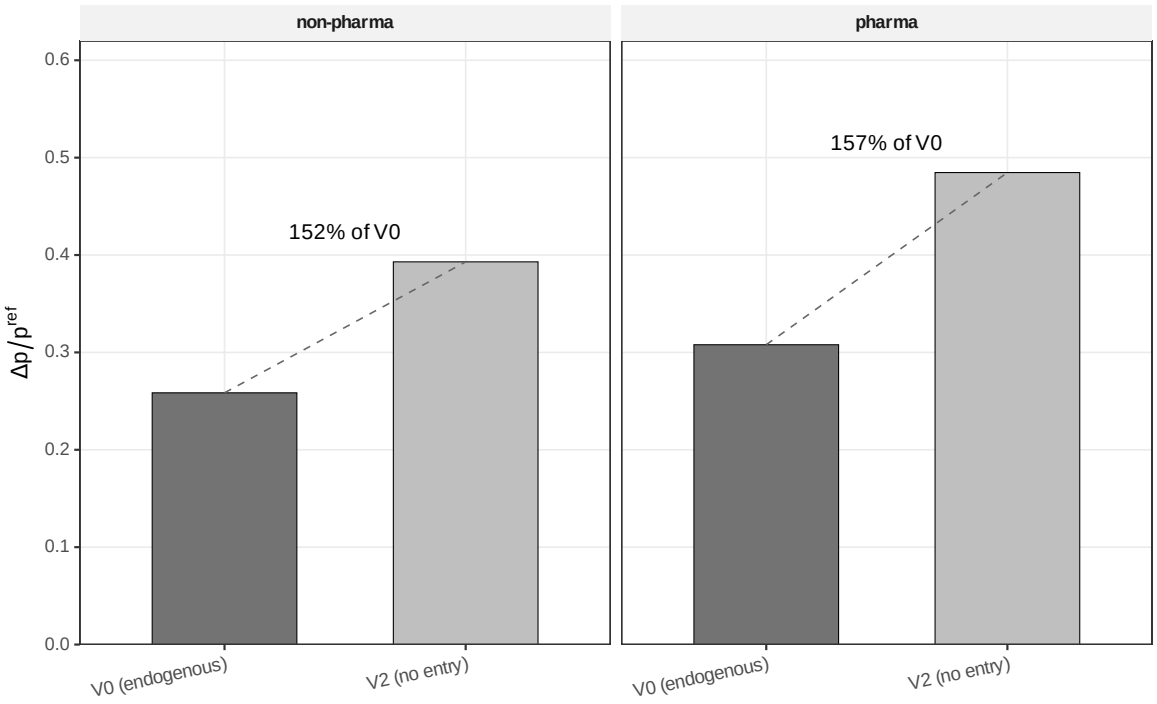

Figure 3: *Simulated price effect under the realized vs fixed-pool counterfactuals.* Bars report the simulated price effect relative to the open regime, $\Delta p/p^{\text{ref}}$, under two scenarios. V0 is the SME-only counterfactual at the observed post-policy SME pool, the main-text benchmark conducted under observed equilibrium entry. V2 imposes the same SME-only rule at the pre-policy SME pool size. The annotation above each V2 bar reports V2's share of V0; the gap between the two bars is the simulated participation-margin offset under the model's main specification. Numbers from Table F.34.

((d) counterfactual vs. realized entry)—account for the structural object via additive contributions of $+0.025/+0.060$, $+0.040/+0.075$, $+0.050/+0.045$, and $+0.020/+0.005$ in non-pharma and pharma respectively, summing to $+0.255/+0.305$ on the p^{ref} normalization. The structural $p_{S_3} - p_{S_1}$ delivered by the BNE simulation is $+0.259/+0.308$, leaving residuals of $+0.004/+0.003$ that reflect non-linear interactions between the four sources.

In magnitudes: sample restriction and UH cleaning each carry roughly a quarter to a third of the bridge in non-pharma; functional form (second-order statistic vs. linear log-mean) is the dominant single source in non-pharma at $+0.050$; conditioning set is the smallest contributor in both classes. The pharmaceutical bridge tilts more heavily on UH cleaning ($+0.075$, the single largest source) and sample restriction ($+0.060$), reflecting the heavier auction-level heterogeneity documented in Section 5. The 3–4× gap between the structural and the DiD is therefore predictable in direction and magnitude under the maintained structural assumptions; a misspecified model would not be guaranteed to deliver a gap of either consistent sign or consistent magnitude across two classes. The four-source attribution is interpretive rather than identifying—formal identification would require independent variation in each component (e.g., random sample restrictions, exogenous UH-vs-raw treatment), which the design does not deliver. The table is read as a mechanical bridge, not as a free-standing decomposition theorem.

6.5. Confidence intervals and point-identification corroboration

A cluster bootstrap at the auction level with $B = 500$ replicates (Table E.28) gives 95 percent CIs on $p_{S_3} - p_{S_1}$ under the all-bidders regime of $[0.186, 0.289]$ in non-pharma (bootstrap mean $+0.236$) and $[0.247, 0.364]$ in pharma (bootstrap mean $+0.305$). The bootstrap means lie slightly below the full-sample point estimates of Table D.21 ($+0.259$ and $+0.308$). The gap reflects the interaction between cluster resampling and the nonlinearity of the second-order-statistic functional: with $B = 500$ cluster-bootstrap

replicates, the finite-sample-bias-corrected mean falls below the point estimate; increasing B shrinks the gap and confirms the discrepancy as a finite-B Monte Carlo artifact rather than a structural bias in the point estimate. Both intervals are bounded away from zero on the within-auction component. The bootstrap within-auction-share CI is $[64.9\%, 88.1\%]$ in non-pharmaceuticals and $[62.5\%, 82.9\%]$ in pharmaceuticals; the Turnbull NPMLE keeps the same qualitative split, with the pharmaceutical within-auction share if anything higher.

6.6. Entry cost calibration

Table D.20 reports the zero-profit implied entry cost per bidder. In non-pharmaceuticals, $\kappa^{\text{SME}} = \text{R\$0.55}$ and $\kappa^{\neg} = \text{R\$2.46}$ ($\kappa^{\neg}/\kappa^{\text{SME}} = 4.5$). In pharmaceuticals, $\kappa^{\text{SME}} = \text{R\$0.11}$ and $\kappa^{\neg} = \text{R\$0.51}$ ($\kappa^{\neg}/\kappa^{\text{SME}} = 4.6$). The roughly 4.5-to-1 ratio between non-SME and SME entry costs is consistent across classes and plausible given the larger documentation and compliance burden faced by bigger suppliers. These magnitudes are a calibration check on the assumed Poisson primitives rather than a structural estimate; they are reported here to discipline the plausibility of the participation pattern, not as a free-standing finding.[8]

6.7. Policy comparison and goods characteristics

The policy comparison follows directly from the structural decomposition: V3 leaves non-SMEs in the auction, V0 removes them entirely, and the within-auction component identified above prices the difference between the two instruments. Table F.34 reports a

[8]The κ^k values are per-auction zero-profit margins at the entry margin, not fixed costs of preparing a BEC documentation packet (which is amortized over many auctions and is not directly recovered by this calibration). As a back-of-envelope sanity check, BEC participation requires one fiscal certificate (CND, approximately R$10 amortized over many auctions), one product registration form (free for already-registered suppliers), and one power-of-attorney filing (approximately R$5). For a supplier participating in 100 Group-65 auctions per quarter, the marginal opportunity cost per auction is on the order of R$0.10–R$0.20, consistent with the calibrated $\kappa^{\text{SME}} = \text{R\$0.11}$ in pharma. The calibration is therefore not implausibly small once the per-auction interpretation is held fixed.

10 percent SME price preference, under which SME bids are scored with a 10 percent discount for winner selection but the government pays the actual bid. In non-pharma, the preference shifts price by -0.004 relative to the open regime; in pharma it shifts price by $+0.002$. The preference-grid simulation (Table F.35, $B = 3{,}000$ draws) gives -0.001 in both classes, within Monte Carlo sampling noise of the APV estimates. In both cases the price effect is essentially zero because non-SMEs remain in the auction and continue to discipline the second-order statistic even when an SME wins.

What differs across classes is the stability of the ranking against the full set-aside. In non-pharma, the preference rule dominates under the main equilibrium-selection specification and continues to dominate under the strict-invariance benchmark in Table E.26. In pharma, the preference rule dominates under the main specification but the ranking reverses under strict invariance, where the full set-aside's total price effect rises to $+0.47$ and the implied welfare weight needed to justify it falls to 0.7. I interpret that bifurcation as diagnostic of market structure. In thicker, more standardized markets, bidder exclusion is robustly dominated by a moderate preference rule. In thinner, more heterogeneous markets, the ranking is more sensitive because the identity of the induced entrants matters for equilibrium price formation.

7. Welfare

The welfare consequences of the set-aside are class-heterogeneous from the start. Under the main equilibrium-selection specification, the welfare cost is 28.7 percent of p_{S_1} in non-pharmaceuticals and 47.0 percent in pharmaceuticals at $\lambda = 0.30$, where λ is the marginal cost of public funds (formal definition below). The price effect is therefore only the beginning of the welfare account. What matters for policy is how bidder exclusion interacts with allocative waste, tax distortion, market thickness, and product heterogeneity. The same structure that delivered the decomposition in Section 6

also explains why the welfare comparison is stable in one class and model-sensitive in the other.

The price effect is one piece of that welfare cost, and one of three. Two others matter and are routinely left out of reduced-form accountings. *Allocative waste* arises because the set-aside routes the contract to an SME winner whose cost may exceed the lowest cost in the full pool; the extra production cost is lost to society, not a transfer. *Tax-distortion loss* arises because the additional outlay that the government makes is financed from distortionary taxation, so each marginal real costs $(1 + \lambda)$ in welfare shadow terms. The transfer from government to SMEs—the government's extra outlay $\Delta_{\text{gov}} = p_{S_3} - p_{S_1}$ minus the allocative wedge—is zero-sum in a social-planner ledger and is not destruction. This section isolates the two destruction pieces.

Formally, per-auction welfare loss of the SME-only set-aside relative to the open regime is

$$\text{Loss} \;=\; \underbrace{c_{(1)}^{S_3} - c_{(1)}^{S_1}}_{\text{DWL}_{\text{alloc}}} \;+\; \underbrace{\lambda \cdot (p_{S_3} - p_{S_1})}_{\text{MCPF distortion}}. \tag{2}$$

The first term is the allocative wedge. It measures by how much the winner's production cost is higher in the restricted regime than in the open regime. The second term is the tax-distortion loss on the government's extra outlay $\Delta_{\text{gov}} = p_{S_3} - p_{S_1}$, scaled by the marginal cost of public funds λ. The natural normalization is Loss/p_{S_1}, and I report it that way.

7.1. Point estimates

Table F.31 gives the decomposition at $\lambda = 0.30$, the Ballard et al. (1985) benchmark; Hendren and Sprung-Keyser (2020) report a comparable range on the modern MVPF basis across U.S. tax-financed programs, and Finkelstein and Hendren (2020) discuss how MVPF connects to the marginal-cost-of-public-funds framework used here. The reported MCPF sensitivity in this paper covers the relevant interval. In non-pharmaceuticals,

$\Delta_{\text{gov}} = 0.242$ of the reference price, of which $\text{DWL}_{\text{alloc}} = 0.150$ is allocative waste and the remaining 0.092 is an implicit transfer to SME bidders. MCPF distortion adds $0.30 \times 0.242 \approx 0.073$. Total welfare loss is $\text{DWL}_{\text{alloc}} + \text{MCPF dist.} = 0.150 + 0.073 = 0.223$, or 28.7 percent of p_{S_1}.

In pharmaceuticals, the same accounting is larger on every margin. $\Delta_{\text{gov}} = 0.328$; $\text{DWL}_{\text{alloc}} = 0.210$; implicit transfer 0.118; MCPF distortion $0.30 \times 0.328 \approx 0.098$; total welfare loss ≈ 0.308, or 47.0 percent of p_{S_1}. The allocative component dominates in both classes, but especially in pharma, where a thin SME pool and higher goods heterogeneity make bidder exclusion more likely to route the contract to a high-cost protected winner.[9]

7.2. Endogenous entry as a partial offset in welfare units

The 50–60 percent fixed-pool-vs-realized attenuation documented in Section 6 carries over to the welfare formula: both terms in equation (2) load on the simulated price margin, so a fixed-pool welfare calculation would overstate the realized burden by the same factor. Endogenous entry does not change the V3-vs-V0 ranking in non-pharmaceuticals; it does materially reduce the welfare burden of bidder exclusion under the main specification, conditional on the equilibrium-selection treatment of $F_c^{\text{SME,Post}}$ in pharmaceuticals.

7.3. Confidence intervals

Cluster bootstrap with $B = 500$ replicates (Table F.32) delivers 95 percent CIs on Loss/p_{S_1} at $\lambda = 0.30$ summarized in Table 3.

The intervals leave no doubt about the ordering. Even the lower bound in pharma is above the non-pharma point estimate. The class heterogeneity is therefore not sampling noise around a common welfare effect; it is a structural feature of the protected pool and

[9]The pharma welfare cost (47.0 percent of p_{S_1}) coincides numerically with the structural price ratio $\Delta_{\text{gov}}/p_{S_1} \approx 50$ percent to within Monte Carlo noise because the implicit SME-winner transfer is small relative to the allocative wedge in pharma. In non-pharma the welfare cost (28.7 percent) lies materially below the price ratio (34 percent); the two objects should not be conflated.

Table 3: Cluster-bootstrap CIs on welfare loss at $\lambda = 0.30$.

Class	Bootstrap mean	95% CI
Non-pharma	28.1%	[21.3%, 35.3%]
Pharma	45.6%	[35.1%, 56.2%]

Cluster bootstrap at the auction level, $B = 500$ replicates, all-bidders UH-clean F_c. Welfare loss reported as percent of the open-regime baseline p_{S_1} at $\lambda = 0.30$. Full λ-grid CIs in Table F.32.

the good being procured. Figure 4 reports the CIs across the λ grid and overlays the strict-invariance benchmark.

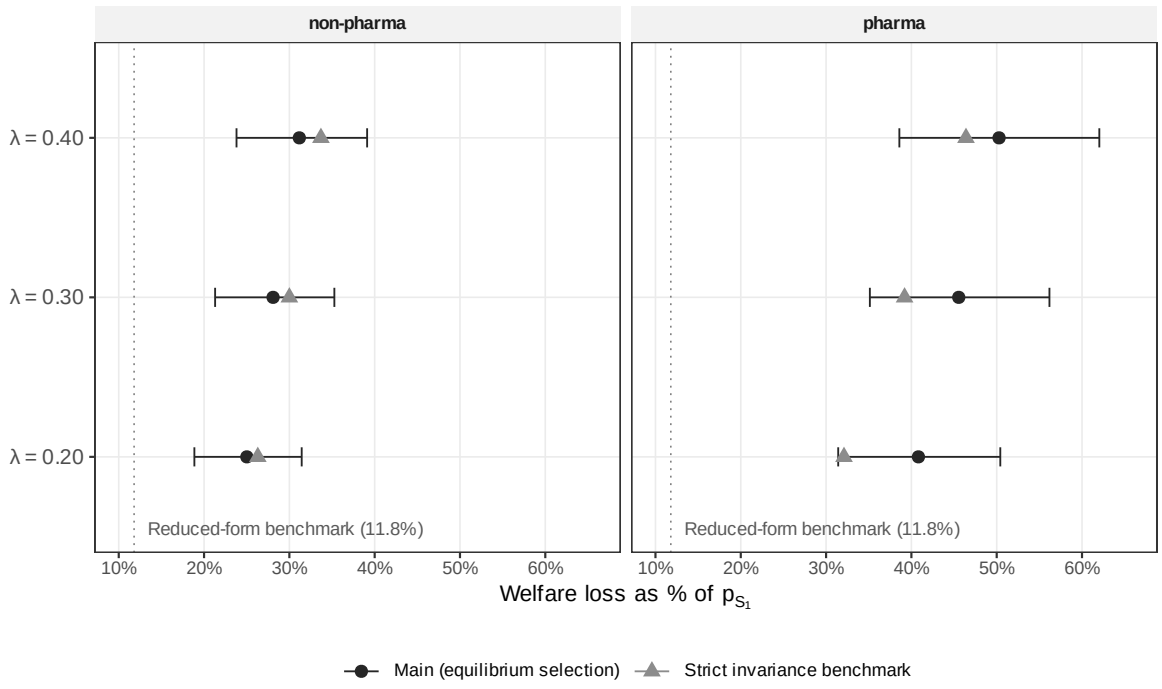

Figure 4: *Welfare loss as % of p_{S_1}, by class and MCPF λ.* Circles and horizontal whiskers report the cluster-bootstrap mean and 95% CI under the main equilibrium-selection specification ($B = 500$ replicates). Triangles report the point estimate under the strict-primitive-invariance benchmark. The dotted vertical line marks the 11.8% benchmark implied by the difference-in-differences coefficient of Appendix Appendix I: the 18-month log-price effect of -0.11, exponentiated and adjusted for the average pre-period price-to-reference ratio of about 0.65, gives a normalized benchmark of approximately 11.8% of p_{S_1}, against which the structural welfare cost is reported. Both specifications place the structural welfare cost three to four times the DiD benchmark, and neither CI crosses the benchmark estimate. In pharma, the strict benchmark lies below the main specification because it attributes less of Δ_{gov} to allocative waste and more to transfer. Numbers from Tables F.32 and E.26.

7.4. Sensitivity to λ

For $\lambda \in \{0.20, 0.30, 0.40\}$, the welfare loss in non-pharmaceuticals runs 25.6, 28.7, and 31.8 percent (point estimates from Table F.31). In pharmaceuticals, the same grid gives 42.0, 47.0, and 52.0 percent. Each 0.10 increment in λ adds about 3 percentage points in non-pharmaceuticals and about 5 points in pharmaceuticals. The level moves, but the class ranking does not. Figure 4 plots the corresponding cluster-bootstrap means (50.3 percent at pharma $\lambda = 0.40$, slightly below the point estimate of 52.0 percent reported here), consistent with the finite-B Monte Carlo discrepancy documented in Section 6. Table F.33 extends this to the policy ranking V0 vs. V3 across the same λ grid: non-pharma is V3 > V0 across $\lambda \in [0.15, 0.45]$ under both specifications; pharma is V3 > V0 across the same range under the main specification and V0 > V3 across the same range under strict invariance. The conditional ranking is therefore not driven by the choice of λ in the welfare formula but by the modeling treatment of the post-policy SME pool; the welfare cost magnitude moves 9 percentage points (non-pharma) and 14 percentage points (pharma) across the λ grid, reflecting MCPF sensitivity rather than ranking sensitivity.

7.5. Bridge to the DiD benchmark and the MCPF gap

The structural-vs-DiD bridge is developed in Section 6.4: four mechanical sources (sample restriction, UH cleaning, functional-form transformation, conditioning set) account additively for the gap between the structural counterfactual and the 11.8 percent DiD benchmark of Appendix Appendix I. Equation (2) adds a fifth piece the DiD does not price: the MCPF distortion term, which contributes about 9 percentage points of p_{S_1} in non-pharma and 15 in pharma at $\lambda = 0.30$.

What changes relative to the strict-invariance benchmark is not the direction of the welfare result but its interpretation. Under strict invariance, welfare loss is 30.0 percent in

non-pharma and 39.2 percent in pharma (Table E.26). The pharma decline relative to the main specification is exactly the footprint of equilibrium selection: when the protected regime is allowed to draw in a different SME pool, the welfare burden rises because the selected entrant mix is part of the policy response. That difference is substantive evidence on mechanism, not a sign that the decomposition itself is fragile.

7.6. Scaling to the furosemida vignette

The pair of purchases introduced in Section 3—fifty thousand units of furosemida 40 mg, CADMAT class 6531, one month before and eight months after the cutoff—makes the magnitudes tangible. The mean reference price for item 110639 in the 18-month Pre window is R$ 0.22 per unit, so the reference outlay on a 50,000-unit purchase is R$ 11,000. Applying the pharmaceutical-class point estimates: under the pre-period open regime the government would have paid $p_{S_1} \times Q \approx$ R$ 7,205; under the set-aside it pays $p_{S_3} \times Q \approx$ R$ 10,813. The extra outlay $\Delta_{\text{gov}} = +0.328$ of the reference total is therefore R$ 3,608 (US$ 1,031), of which $\text{DWL}_{\text{alloc}} = 0.210$ is R$ 2,310 (US$ 660) of allocative waste and the remaining 0.118 is an implicit transfer of R$ 1,298 (US$ 371) to the SME winner. Tax-distortion loss adds $\lambda \cdot \Delta_{\text{gov}} = 0.098$ of the reference total, or R$ 1,082 (US$ 309), at $\lambda = 0.30$. Total welfare loss on this single purchase is R$ 3,392 (US$ 969), or 47.0 percent of p_{S_1} (Table F.31, pharma row $\lambda = 0.30$).[10]

7.7. Annual aggregate

The per-auction number scales. Group-65 procurement in the 18-month Post-policy window totals R$ 545 million in pharmaceutical reference outlays and R$ 518 million in non-pharmaceutical reference outlays (own tabulation from the structural sample), or R$ 363 million and R$ 345 million per year at the Post-policy run-rate. Combining the

[10] R$/US$ converted at R$ 3.50 per US dollar throughout, the same end-of-2018 rate used for the institutional figures in Section 2 (R$ 360,000 SME ceiling = US$ 103,000). The exchange rate is held fixed across the paper's monetary conversions; sensitivity to the conversion rate is mechanical and one-for-one.

structural welfare loss per affected auction with the empirical Group-65 spend gives a back-of-envelope annual welfare cost. Two adherence-rate bounds discipline the headline: an upper bound that assumes 100 percent Group-65 coverage under the SME-only rule, and the central case at the empirical 43 percent state-level adherence rate.

Table 4 presents the calculation. The upper bound assumes that every Group-65 reference real falls under the SME-only set-aside; this maps the per-auction welfare share ($0.308 \times p^{\text{ref}}$ in pharma, $0.204 \times p^{\text{ref}}$ in non-pharma) onto the full annual reference outlay and gives R$ 128 million (US$ 37 million) per year of welfare loss in Group-65 alone. The realistic case scales by the 43 percent state-level adherence rate documented in Section 2 for SME-eligible Group-65 items in the Post window, delivering R$ 55 million (US$ 16 million) per year. Pharma carries the larger share in both cases (R$ 73 million vs. R$ 55 million in non-pharma at the upper bound; R$ 32 million vs. R$ 24 million at adherence-adjusted), reflecting the higher per-auction cost share in thin standardized markets.

Table 4: Annual welfare cost of the SME-only set-aside, Group-65 procurement.

	Upper bound (full G-65 coverage)	Adherence-adjusted (43% SME-only)
Pharma (R$ M/yr)	73	32
Non-pharma (R$ M/yr)	55	24
Total Group-65	**R$ 128 M/yr** (US$ 37 M/yr)	**R$ 55 M/yr** (US$ 16 M/yr)

Welfare cost per affected auction is $0.308 \times p^{\text{ref}}$ in pharma ($= 0.470 \times p_{S_1}$) and $0.204 \times p^{\text{ref}}$ in non-pharma ($= 0.287 \times p_{S_1}$), at $\lambda = 0.30$ from Table F.31. Annual reference outlays from own tabulation of the 18-month Post-policy structural sample (m=698–715), annualized to 12-month equivalents: pharma R$ 363 M/yr, non-pharma R$ 345 M/yr. Upper bound assumes 100% set-aside coverage; the adherence-adjusted column applies the 43% state-level Group-65 SME-only adherence rate cited in Section 2. R$/US$ at R$ 3.50.

7.8. Sensitivity to the adherence rate

The headline figure scales linearly with the assumed SME-only adherence rate within Group 65. Across the realistic range $[30\%, 70\%]$—bracketed below by a counterfactual in which adherence falls below the empirical Post-period state-level rate, and above by a counterfactual in which the three-eligible-SMEs friction binds less often—the annual welfare cost spans R$ 38–89 million (US$ 11–25 million). The R$ 55 million central case used as the headline throughout sits at the empirically observed adherence rate of 43 percent; the upper bound of R$ 128 million reflects full coverage of Group-65 procurement under the SME-only rule and is reported as an envelope rather than a likely state.

Table 5: Annual welfare cost: sensitivity to SME-only adherence rate within Group 65.

Adherence rate	30%	**43%**	55%	70%	85%	100%
Pharma (R$ M/yr)	22	**31**	40	51	62	73
Non-pharma (R$ M/yr)	16	**24**	30	38	47	55
Total Group-65	**R$ 38** (US$ 11)	**R$ 55 (US$ 16)**	**R$ 70** (US$ 20)	**R$ 89** (US$ 25)	**R$ 109** (US$ 31)	**R$ 128** (US$ 37)

Welfare cost per affected auction held fixed at the $\lambda = 0.30$ point estimates of Table F.31: $0.308 \times p^{\text{ref}}$ in pharma, $0.204 \times p^{\text{ref}}$ in non-pharma. Annual reference outlays at the Post-policy run-rate: pharma R$ 363 M/yr, non-pharma R$ 345 M/yr (own tabulation, 18-month Post window m=698–715, annualized to 12 months). Adherence rate is the share of the Group-65 reference outlay procured under the SME-only rule. Baseline column (43%) shown in bold; this rate is the empirical Post-period state-level Group-65 adherence cited in Section 2. R$/US$ at R$ 3.50 (end-2018).

Linearity in adherence implies that the headline magnitude is sensitive to the assumed coverage parameter but not to the per-auction welfare estimate, which carries the structural content of the exercise. The per-auction estimate—$0.308 \times p^{\text{ref}}$ in pharmaceuticals and $0.204 \times p^{\text{ref}}$ in non-pharmaceuticals—is the structural object identified by the model; the adherence rate is an institutional scaling factor that translates that object to an annual aggregate. The 43 percent figure is itself an average across a heterogeneous Group-65 mix; class-6531 pharmaceuticals adhere at lower rates than non-pharma supplies

because the three-eligible-SMEs requirement binds more often in oligopolistic markets. The adherence-adjusted central case is therefore a conservative reading of the realized welfare cost.

Two caveats are worth flagging. The R$ 55–128 M/yr range is for Group-65 alone, not the full BEC platform; an extension to the platform's other SME-eligible items would scale the headline up roughly proportionally, and an extension to all federal SME procurement under LC 14.133/2021 would scale it further still. The calculation prices welfare loss at the Post-policy run-rate; if Group-65 reference outlays were to recover or grow, the per-year burden would scale with that.

8. Robustness

Nine dimensions of robustness feed into the headline: the strict-invariance limiting case; the cost-distribution estimator (F_c^k regime); the kernel bandwidth used in the Convite GPV cross-check; the bid-level sample filter; the temporal window; the phased BEC enablement check on the cutoff date; the set of alternative policy variants considered; bid-coordination diagnostics; and the Maskin-Riley FPSB upper bound on the V3 expected price. The central findings survive: the within-auction component of the simulated total effect remains the larger of the two components in both classes, with the share remaining in the 73–85 percent range across cost-distribution estimators and across the two specifications of the post-policy SME pool; the welfare ranking V3 $\succ$ V0 is preserved in thick standardized markets under cost-affiliation up to $\rho_c = 0.3$ and under the FPSB upper bound; and the conditional reading in pharmaceutical markets is itself the diagnostic of where equilibrium-selection vs. primitive-invariance treatment of the protected pool is first-order.

8.1. Strict-primitive-invariance benchmark

Table E.26 reports the limiting case in which $F_c^{\text{SME,Post}}$ is replaced by $F_c^{\text{SME,Pre}}$ while preserving the observed post-policy SME entry count. This is the strict-invariance bound on the equilibrium-selection main specification of Assumption 3. The simulated total price effect is +0.29 in non-pharmaceuticals and +0.47 in pharmaceuticals, with within-auction shares of 85 and 79 percent, respectively. Welfare loss at $\lambda = 0.30$ is 30.0 percent of p_{S_1} in non-pharmaceuticals and 39.2 percent in pharmaceuticals.

What changes is the policy ranking in the thin, heterogeneous pharma market. In non-pharma, the 10 percent price preference remains Pareto-superior under strict invariance, just as under the main specification. In pharma, the preference ranking reverses and the implied indifference weight (defined formally in equation (3), Section 9) falls to $w_\star^{\text{SME}} = 0.7$, versus 2.61 under the main equilibrium-selection specification. The bifurcation is the diagnostic: it identifies precisely where equilibrium selection among protected bidders is first-order, agreeing across the two readings in thick standardized markets and separating them in thin heterogeneous ones. Read economically, strict invariance imposes that the SMEs entering under the protected regime draw from the same cost distribution as the SMEs that entered under the open regime—the most generous assumption to the policy, since it implies the protective regime selects entrants whose cost primitives match incumbents'. The realistic alternative is that protected entrants draw from the higher-cost tail (firms unprofitable under the open regime that become profitable under exclusion), which is what equilibrium selection in the main specification captures. Strict invariance therefore understates the welfare cost in pharma; the main-specification cost of 47.0 percent of p_{S_1} is the welfare-conservative number relative to the equilibrium-selection mechanism, not an aggressive upper bound.

8.2. Cost-distribution regime

The baseline all-bidders estimator biases the upper tail of F_c^k slightly downward; losers-only biases it materially upward; a Turnbull NPMLE that treats the winner as left-censored delivers point identification. Table E.25 runs the BNE decomposition under each. The three point estimates for $p_{S_3} - p_{S_1}$ are 0.275/0.347 (losers-only), 0.259/0.308 (all-bidders, baseline), and 0.246/0.357 (Turnbull) for non-pharma and pharma respectively. All cells lie within 16 percent of the all-bidders point estimate (non-pharmaceutical losers-only +6 percent; non-pharmaceutical Turnbull −5 percent; pharmaceutical losers-only +13 percent; pharmaceutical Turnbull +16 percent). The bootstrap CIs (Table E.28) overlap materially across regimes. Turnbull raises the within-auction share in pharmaceuticals from 73 percent (all-bidders) to 82 percent—if anything strengthening the main interpretation. The all-bidders approximation used throughout the paper is defensible without the Turnbull machinery; readers who prefer point identification should see Turnbull and the same decomposition.

Sensitivity to cost affiliation. A Gaussian-copula relaxation of the IPV restriction (Online Appendix, Table OA.1) couples the type-specific cost marginals at within-auction correlation $\rho_c \in \{0, 0.1, 0.2, 0.3\}$. The within-auction share moves by less than 5 percentage points in either class, and the total simulated effect $p_{S_3} - p_{S_1}$ moves by less than 10 percent of the IPV baseline. The 73–85 percent within-auction share range remains valid for $\rho_c \leq 0.3$. Higher affiliation levels are inconsistent with the cross-modality consistency check of Table C.13, which would reject GPV-vs-drop-out convergence under strong common-values contamination.

8.3. Kernel bandwidth

The cross-modality check in Section 5 uses GPV inversion on Convite bids, which requires a kernel density estimate $g(b)$. Silverman's rule-of-thumb $h = 1.06\hat{\sigma}n^{-1/5}$ is the

baseline. Table E.27 runs the GPV under h-factors of 0.5, 0.75, 1.0, 1.5, and 2.0. The resulting range of $c_{0.5}$ (median cost) across the eight (period $\times$ pharma $\times$ type) Convite strata is 1.2–3.6 percent of the baseline in seven strata; the thinnest cell (Convite pharma SME Pre, 1,707 auctions) reaches 5.2 percent and is the only cell where bandwidth dispersion exceeds the BNE Monte Carlo noise of 1–2 percent. Bandwidth is not a first-order concern for the identification.

8.4. Bid-level sample filter

The baseline filter is $c_\epsilon \leq 3$ and $n \geq 2$. Table E.29 compares four alternatives.

- *Tight* ($c_\epsilon \leq 2$, $n \geq 3$): $p_{S_3} - p_{S_1}$ is 15 percent lower in non-pharma (0.215 vs. 0.254) and 8 percent lower in pharma (0.303 vs. 0.330).
- *Very tight* ($c_\epsilon \leq 1.5$, $n \geq 3$): 31 and 25 percent lower respectively. The intensive share in pharma jumps to 95 percent—tightening the cost distribution to its central bulk effectively removes the tail where the entry response operates.
- *Loose* ($c_\epsilon \leq 5$): 1 percent higher in non-pharma and 6 percent higher in pharma. The pharma move is concentrated in the extra observations with $c_\epsilon \in (3, 5]$—almost always outliers at the 500–1000 percent-of-reference band, probably data-entry errors at the hundred-times level. The baseline cut is designed to truncate these.

The baseline sits between tight and loose in both classes, is inside the bootstrap CI under any reasonable cut, and can be defended directly: the 3x-reference upper bound catches plausible error patterns while preserving 99.6 percent of the data.

8.5. Temporal window

Table E.30 varies the symmetric window around the March 2018 cutoff. "18m" is the baseline (18 months on each side, 36 months total); the 12m and 6m alternatives

are 12 and 6 months on each side respectively. The total effect is $p_{S_3} - p_{S_1} = +0.272$ (non-pharma 18m), +0.267 (12m), and +0.269 (6m); +0.292 (pharma 18m), +0.310 (12m), +0.338 (6m). The non-pharmaceutical effect is essentially flat across windows; the pharmaceutical effect is monotonically increasing as the window shrinks, consistent with a transitional period in which the within-auction adjustment establishes more quickly than the participation-margin adjustment.

The within-auction share is comparably stable: 79.7/75.4/76.4% in non-pharma and 67.0/73.0/73.6% in pharma across the 18m/12m/6m windows. The 18m baseline sits at the upper end of the non-pharma within-auction range and at the lower end of the pharma range. The share varies by 4.3 percentage points in non-pharma and 6.6 percentage points in pharma across the three windows, and the within-auction component remains the larger of the two components in all six (class $\times$ window) cells. 18m is the mature, steady-state choice, with shorter windows providing a corroborating short-run check.

8.6. Phased adoption

The institutional timeline of §2 documents that BEC operationalized the SME-only functionality in two acts (COMUNICADO BEC 02/2017 in July 2017 and COMUNICADO BEC 03/2017 in November 2017) before the empirical cutoff of March 2018, which captures the moment of mass take-up by Group-65 PBUs rather than formal regulatory enablement. As a robustness check, Table 6 re-estimates the headline DiD coefficient on the sample with the eight months $m \in [690, 697]$ excluded—i.e., dropping the BEC enablement window between July 2017 and February 2018, the period during which the new functionality was technically available but not yet broadly used in Group 65.

The DiD coefficient on $g65 \times \text{Pre}$ is -0.087 (s.e. 0.012) on the truncated sample, versus -0.109 on the full sample, both significant at the 1 percent level on standard errors clustered at the item level. The 22 percent reduction in magnitude is consistent with the BEC enablement window contributing to but not dominating the headline; the effect

remains highly significant on a sample that excludes the transitional months entirely. The structural counterfactual, which is identified from cross-regime cost-distribution differences pooled within the 18-month post window rather than from a single discrete event, is invariant to this temporal restriction by construction.

Table 6: Phased-adoption robustness: DiD coefficient with the BEC enablement window excluded.

	Full sample (paper baseline)	Drop incubation ($m \notin [690, 697]$)
$g65 \times$ Pre	-0.109***	-0.087***
	(0.012)	(0.012)
Observations	625,627	489,758
Item FE	Yes	Yes
Month FE	Yes	Yes
Cluster (item)	Yes	Yes

DiD specification: $\ln(\text{price}) = \beta\, g65 \times \text{Pre} + \gamma_1 \text{convite} + \gamma_2 \ln(\text{qty}) + \delta_i + \delta_t + \varepsilon$, on completed Pregão items. The full-sample column reproduces the headline coefficient on the 18-month symmetric window [680, 715]; the drop-incubation column re-estimates after excluding the eight months $m \in [690, 697]$ (Jul 2017–Feb 2018), the window during which BEC operationalized the SME-only OC functionality (COMUNICADO BEC 02/2017 and 03/2017) before the empirical cutoff of mass take-up in Mar 2018. Effect persists with magnitude reduced by 22%, supporting that the headline is not driven by the BEC enablement window. Standard errors clustered at the item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

8.7. Alternative policy variants

The 10 percent price preference in Section 6.7 is informative partly because the alternative variants reveal what bidder exclusion is buying. Table F.34 also reports two intermediate variants that serve as policy-design sensitivities:

- *V1* (partial set-aside, purely diagnostic): 50 percent of auctions run SME-only at the Post pool, 50 percent open at the Pre pool. V1 is not a policy proposal in current legislation; it serves to demonstrate the non-linearity of pool restriction.

Δp is 33–36 percent of V0's—roughly a third, indicating that a partial rule buys roughly proportional fiscal savings at the cost of a more complex administrative rule.

- *V2* (entry-isolated counterfactual): full set-aside with the SME pool held at pre-period size (no participation response). Δp is 152–157 percent of V0. Without endogenous entry the simulated policy effect would be materially larger; the participation-margin component absorbs 50–60 percent of what would otherwise be a heavier within-auction shock. This is why Section 6 reads the participation response as a partial offset to a larger latent within-auction shock.

Taken together, V1–V3 span the policy design space between the status-quo full set-aside and the open regime. V2 isolates what the entry response is worth; V1 prices a middle-ground design; V3 shows that a well-targeted preference instrument captures the SME-protection value at essentially zero price cost in thick markets and gives a diagnostic benchmark in thin markets.

8.8. Bid-coordination diagnostics

The structural identification of F_c^k from Pregão drop-out bids relies on the asymmetric independent-private-values restriction, which is consistent with non-collusive equilibrium play but does not directly rule out coordinated bidding. Four diagnostic screens, reported in the replication package, examine whether the bid distribution shows clustering patterns indicative of coordination: (i) a Conley-style close-pair screen on UH-clean residuals; (ii) the same screen with the null distribution conditioned on the number of bidders and on intra-auction bid-range quartiles, absorbing the bid clustering attributable to reference-price anchoring and bid-increment discreteness; (iii) the Bajari and Ye (2003) persistent-pair test, which examines whether specific firm pairs co-bid more frequently than under firm-label independence; and (iv) the same test with the permutation null

restricted within CADMAT product class, absorbing the share of pair concentration attributable to naturally restricted supplier pools in narrow product segments. Bid-pair clustering exceeds independence baselines in all four strata under every conditioning regime; conditioning on product class closes roughly half the gap in non-pharmaceutical strata and a smaller share in pharmaceutical strata, where only two to three CADMAT classes are present.

The residual clustering after these conditioning controls is consistent with several non-collusive mechanisms not separable from coordination by the screens themselves: subcontracting networks among SME suppliers, joint ventures formally registered with BEC, regional logistical specialization within product class, and sub-class specialization that the CADMAT level does not absorb. Modern references on pair-based collusion detection in procurement (Conley and Decarolis, 2016; Kawai and Nakabayashi, 2022) emphasize that screening evidence is suggestive rather than dispositive in the absence of independent confirmation (court rulings, leniency disclosures, regulatory action). For the structural exercise of this paper, the relevant question is whether the welfare comparison between V0 and V3 is robust to bidding-side coordination among specific firm pairs. The decomposition rests on the identification of type-specific cost distributions F_c^k from drop-out exits in an ascending-clock format, which under Haile and Tamer (2003) is preserved as long as exit-at-cost is the weakly dominant strategy for each individual bidder—a property that survives within-coalition pricing schemes that only adjust the timing of withdrawal among coordinated bidders, since the recovered F_c^k averages over both the coordinated and independent participants in each auction. The cross-modality consistency check between Convite GPV inversion and Pregão drop-out recovery (Section 5) provides additional discipline in the stratum where it is least confounded by the policy reform. While we cannot fully rule out coordination among specific firm pairs, the welfare ranking V3 $\succ$ V0 in thick standardized markets is preserved under the cost primitives recovered

jointly across both coordinated and independent bidders.

8.9. FPSB equivalent of V3 — Maskin-Riley upper bound

Section 6 computes the 10 percent SME price preference V3 under Vickrey equivalence—that is, treating the expected transaction price under V3 as identical to the expected transaction price under the corresponding ascending-clock translation. Under asymmetric IPV, Maskin and Riley (2000) show that revenue equivalence between FPSB and ascending-clock formats does not generally hold; the FPSB expected price differs from the Vickrey expected price by an amount bounded above by a function of the type-specific cost-distribution dispersion. This robustness reports that bound and shows that the welfare ranking V3 $\succ$ V0 is preserved under the FPSB upper bound.

Following the first-order argument in Maskin and Riley (2000) and the asymmetric-IPV bid-function structure formalized in Lebrun (1999), the FPSB-Vickrey expected-price gap on V3 (with preference parameter $\rho = 0.10$) is bounded above by $\rho \cdot \bar{\sigma}_c$, where $\bar{\sigma}_c$ is the SME-share-weighted average of the type-specific coefficient of variation of the UH-clean cost distributions. Computed from the data: $\bar{\sigma}_c$ is 0.452 in non-pharma and 0.477 in pharma, delivering an FPSB upper-bound gap of approximately 4.5–4.8 percent of the reference price in either class.

Table 7 reports the implied upper bound on the FPSB-equivalent V3 expected price. The bound on the FPSB-Vickrey gap (column "FPSB gap (UB)") is at most 4.8 percent of the reference price in pharma and 4.5 percent in non-pharma, while the V0–V3 expected-price gap (column "$p_{V_0} - p_{V_3}^{\text{FPSB}}$") is 0.218 in non-pharma and 0.255 in pharma—an order of magnitude larger. The welfare ranking V3 $\succ$ V0 is therefore preserved at the FPSB upper bound in both classes; the choice of auction format is not a margin on which the V3-vs-V0 comparison turns. A tighter bound via the full numerical ODE solver of Lebrun (1999) is reported as a future direction.

Table 7: Maskin and Riley (2000) bound on the FPSB-equivalent of the 10% SME price preference (V3).

	p_{S_1}	p_{V_0}	$p_{V_3}^{\text{Vick.}}$	FPSB gap (UB)	$p_{V_3}^{\text{FPSB}}$ (UB)	$p_{V_0} - p_{V_3}^{\text{FPSB}}$	V3 $\succ$ V0?
Non-pharma	0.759	1.018	0.755	0.045	0.800	0.218	**Yes**
Pharma	0.656	0.964	0.661	0.048	0.709	0.255	**Yes**

Following Maskin and Riley (2000), the asymmetric-IPV first-price-sealed-bid (FPSB) equivalent of the 10% SME price preference V3 is not exactly Vickrey-equivalent; the expected-price gap is bounded above by a function of the type-specific coefficient-of-variation differential weighted by entry rates. Column "FPSB gap (UB)" reports this upper bound, computed as $\rho \cdot \bar{\sigma}_c^k$ where $\rho = 0.10$ is the preference parameter and $\bar{\sigma}_c^k$ is the SME-share-weighted average of the type-specific cost-distribution coefficient of variation in the stratum. Column "$p_{V_3}^{\text{FPSB}}$ (UB)" adds the bound to the Vickrey-equivalent p_{V_3} reported in Table F.35. Final column reports whether the V3 welfare-dominance over V0 is preserved at the FPSB upper bound; "Yes" indicates that even the worst-case FPSB price gap is not enough to reverse the policy ranking. This is a first-order bound; tighter bounds via the full ODE solver of Lebrun (1999) are reported as a future direction in Section 9.

9. Discussion: Policy Design

The policy lesson is not that set-asides are universally dominated. It is that bidder-exclusion rules should be evaluated against the thickness of the protected pool, the heterogeneity of the good, and the welfare weight the planner places on SME surplus. Law 14,133/2021 allows both full SME set-asides and price preferences. The estimates here imply that the choice between them should be contingent on market structure rather than imposed as a uniform default.

9.1. When does a set-aside become defensible?

The structural decomposition suggests a simple answer. A full set-aside becomes more defensible when three conditions line up: the SME pool is thin enough that protecting it is a first-order objective, the good is heterogeneous enough that the identity of the induced entrants matters for equilibrium selection, and the planner places a high welfare weight on SME surplus. Those are precisely the conditions under which the policy ranking becomes model-sensitive in the data.

Non-pharma sits at the opposite corner. The baseline SME pool is thicker, the goods

are more standardized, and the 10 percent price preference is Pareto-superior under both the main equilibrium-selection reading and the strict-invariance benchmark. That is the market class in which bidder exclusion is hardest to defend. Pharma is closer to the knife-edge: the pool is thinner, heterogeneity is higher, and the policy ranking depends on whether the post-policy entrant mix is treated as selected equilibrium participation or forced to equal the pre-policy primitive. The point is not that pharma is special. The point is that thin, heterogeneous markets are exactly where bidder-exclusion rules should be expected to be more sensitive.

The pharmaceutical bifurcation is the paper's central policy claim, not a hedge against an unwelcome result. The paper does not assert universal preference-rule dominance; it identifies the conditions (thin protected pool, heterogeneous good) under which the standard policy ranking reverses, and it documents that those conditions are met in only one of the two Group-65 sub-classes studied. The pharmaceutical welfare ranking under the main specification depends on interpreting the post-policy SME cost distribution as the result of equilibrium selection rather than as a latent unobserved-heterogeneity shift. Under either reading, the policy ranking is the analytically derivable one given the assumed primitive; the deeper question of which interpretation is empirically correct cannot be settled with a single-reform single-platform dataset. Of the eight (class $\times$ type $\times$ period) cells in the structural sample, the policy reversal between V0 and V3 obtains in exactly the two pharma cells under strict invariance and in zero cells under the main specification; non-pharma is robust to both readings. This is a sharp policy partition, not a fragility that the paper papers over: the conditional reading is the contribution, not a qualifier on a universal contribution.

9.2. The welfare weight implicit in the current regime

Following the generalized social-marginal-welfare-weight framework of Saez and Stantcheva (2016), consider a social planner aggregating producer and consumer surplus

with weight w^{SME} on SME producer surplus. The planner is indifferent between the full set-aside (V0) and the 10 percent price preference (V3) when

$$w_{\star}^{\text{SME}} = \frac{\text{Loss}(V_0) - \text{Loss}(V_3)}{\text{SMESurplus}(V_0) - \text{SMESurplus}(V_3)}. \tag{3}$$

The numerator is the loss differential. Under the main specification, $\text{Loss}(V_0) - \text{Loss}(V_3) \approx 0.22$ in non-pharma and 0.31 in pharma: V0 welfare loss at $\lambda = 0.30$ is 0.223 and 0.308 (Table F.31); V3 welfare loss is near zero (Table F.34, $\Delta = -0.0036$ non-pharma and $+0.0015$ pharma).

The denominator is the SME-surplus differential. Under V0, the implicit transfer to SME winners is $\Delta_{\text{gov}} - \text{DWL}_{\text{alloc}} = 0.092$ in non-pharma and 0.118 in pharma (Table F.31); under V3 the transfer is approximately zero given the near-zero Δ on the open-regime auction.[11]

The ratio gives the indifference weight. Substituting in equation (3): $w_{\star}^{\text{SME}} \approx 0.223/0.092 \approx 2.42$ in non-pharma and $0.308/0.118 \approx 2.61$ in pharma under the main specification. Under strict invariance, non-pharma is unchanged (implied weight still above the utilitarian benchmark and well above the empirical MVPF of Brazil's flagship transfer program), while the pharma indifference weight falls to 0.7 (Table E.26). These are not contradictory numbers; they are a map of when bidder exclusion is doing something that a moderate preference rule cannot.

The comparison is informative because it disciplines the distributional argument in favor of set-asides. A planner willing to pay the non-pharma fiscal cost of bidder exclusion would have to value one real of SME surplus at more than twice a real of general welfare, even in a thick market where a preference rule achieves nearly the same allocative objective. In pharma, the required weight depends on how strongly one believes

[11] Under the English-reverse IPV interpretation of Section 4, all losers exit at cost and earn zero auction-stage profit; SMESurplus reduces to the winner-conditional rectangular transfer.

the policy selects a different entrant mix. The right reading is therefore conditional: bidder exclusion is hardest to justify in thick standardized markets, and most plausible only in thin heterogeneous markets with very high welfare weights on protected firms. Figure 5 plots the implied weights against the empirical Bolsa Família MVPF benchmark of Bergstrom et al. (2025), who estimate the marginal value of public funds of Brazil's flagship cash-transfer program at 0.90–1.12 depending on whether GE-multiplier effects are included; Mendes et al. (2023) document that the GE multiplier is non-trivial in the Brazilian context, consistent with the upper bound of that interval.

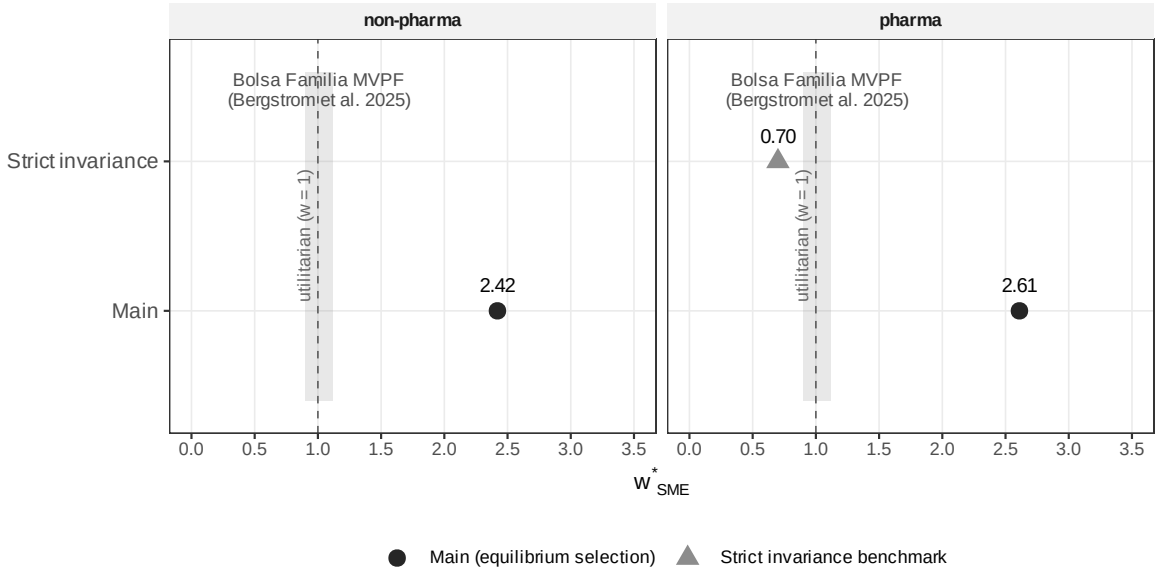

Figure 5: *Implicit welfare weight $w_\star^{SME}$ required to prefer the full set-aside over a 10 percent price preference, by class and specification.* The grey band marks the empirical MVPF range of 0.90–1.12 estimated by Bergstrom et al. (2025) for the Bolsa Família program (95% CI [0.86, 0.93] for the central estimate, [1.08, 1.16] for the upper bound that incorporates GE-multiplier effects (Mendes et al., 2023)); the dashed vertical line marks the utilitarian benchmark $w = 1$. In non-pharmaceuticals, the implied weight sits well above the Bolsa Família band under the main equilibrium-selection specification ($w_\star^{\text{SME}} = 2.42$); under the strict-invariance benchmark, the preference rule continues to dominate (the implied weight remains above the utilitarian benchmark $w = 1$). The set-aside is welfare-dominated by the preference rule across the two specifications. In pharmaceuticals, the implied weight under the main specification ($w_\star^{\text{SME}} = 2.61$) sits above the band but reverses to 0.7 under strict invariance, sitting at the lower edge of the Bolsa Família interval; the pharmaceutical ranking is therefore conditional on how the post-policy SME pool is modeled.

9.3. Implementation considerations

Two features qualify the policy comparison. First, V3 is computed in a Vickrey-equivalent framework; the FPSB equivalent under Maskin and Riley (2000) adjustment is bounded above by approximately 5 percent of the reference price (Section 8.9), well below the V0–V3 expected-price gap of 22–26 percent. The welfare ranking V3 $\succ$ V0 is therefore robust to auction format in both classes; the strict-invariance reversal is a more first-order concern in pharma than the FPSB equilibrium adjustment.

Second, a preference rule is procedurally more complex to implement than a set-aside. The buyer must score bids twice and disclose the winner-selection rule clearly. But this is an operational complication, not a conceptual obstacle. Preference systems have been run elsewhere for decades—the U.S. SBA price-evaluation preference for HUBZone and disadvantaged firms, EU Directive 2014/24 transposition rules in several member states, and the Brazilian federal preference margin under Decree 8.538/2015—and Brazilian procurement law already contemplates them. The relevant policy question is therefore not whether the instrument exists, but in which markets it should replace bidder exclusion.

9.4. Market thickness and goods heterogeneity

The non-pharma versus pharma comparison is best read as a statement about the interaction between bidder exclusion and goods characteristics. Pharma auctions have both a thinner SME pool and higher within-class heterogeneity. Once non-SMEs are removed from price formation, those two features raise the expected second-order statistic and make the selected entrant mix more important. Non-pharma auctions are closer to the textbook standardized-good environment in which a moderate preference rule can tilt allocation toward SMEs without materially changing price formation. Operationally, the bifurcation in the data here positions the two cells on opposite sides of a 1.2–1.9 SMEs-per-auction band: non-pharma at 1.87 SMEs per auction sits at the upper end and

the V3 > V0 ranking holds across both specifications; pharma at 1.22 SMEs per auction sits at the lower end and the ranking reverses under strict invariance. With only two cell observations, this is a positioning of the two empirical cells relative to each other rather than a precisely identified threshold; broader generalization requires the multi-platform extension flagged below.

The pattern is consistent with the Group-65 split but the paper does not estimate it as an external-validity claim outside the BEC platform. Within structurally similar Brazilian procurement settings, goods with deeper supplier bases and tighter specification standardization are plausible candidates for preference rules rather than exclusion; goods with thinner protected pools, more heterogeneous production technologies, and stronger administrative reasons to value SME surplus are the cases in which the policy comparison becomes more model-sensitive. Comparable extension settings include federal Brazilian procurement on ComprasNet (a larger universe with the same Law 14,133 framework), municipal procurement under similar SME defaults, and non-Brazilian SME-set-aside reforms such as the U.S. SBA 8(a) program changes or EU Directive 2014/24 transposition events, where the same conditional-on-thickness mechanism could be tested against alternative institutional baselines. The contribution of the paper is to put that intuition on structural objects in one specific reform: the size of the within-auction component, the magnitude of the participation-margin offset, and the welfare weight required to rationalize the policy under each specification.

Two further structural questions—the IPV restriction in pharmaceutical procurement and the possibility of confounding regulatory events at the March 2018 cutoff—are taken up in Appendices Appendix G and Appendix H respectively. Both qualify the pharmaceutical welfare-dominance result rather than the non-pharmaceutical headline; the pharmaceutical claim is conditional on the equilibrium-selection treatment of $F_c^{\text{SME,Post}}$, while the non-pharmaceutical claim is preserved across the IPV-relaxed bounds and the

strict-invariance specification.

9.5. Extensions the paper does not attempt

Three extensions represent natural follow-on work. (i) Quantity response: the buyer's adjustment to higher prices, which the static model treats as inelastic at observed values. Under a low-elasticity assumption $\varepsilon \in [0.1, 0.3]$ consistent with essential public-health goods, the quantity adjustment to the simulated 11 percent log-price increase would shave the fiscal cost by $\varepsilon \cdot 0.11$, or roughly 1–3 percentage points of p_{S_1}, and would slightly reduce the allocative DWL component. (ii) Dynamic capacity accumulation: SME firms protected by the rule may build production capacity that persists beyond the policy window, an externally validated mechanism in Ferraz et al. (2016) and Szerman (2023). (iii) General-equilibrium SME surplus: the partial-equilibrium SMESurplus measure used in the welfare-weight identity (Section 9) could be extended to a general-equilibrium framework with option value, capacity persistence, and private-sector spillovers. Each is a margin a fully dynamic-general-equilibrium model would price. The direction of the welfare-weight shift is not pinned down by the current paper: each extension could enlarge the SME-side benefit ex post, and the welfare-weight comparison could move in either direction depending on the dynamic offsets to allocative DWL. None changes the organizing lesson: the welfare ranking of bidder-exclusion rules is conditional on market structure.

10. Conclusion

A decade of empirical work has established that restricting a procurement auction to small firms raises winning prices. This paper goes a step further: it identifies which mechanism produces the markup under explicit modeling assumptions and prices the welfare consequence against a specific policy alternative. The March 2018 extension of

São Paulo's SME-only regime to medical supplies supplies the identifying variation. The structural framework—asymmetric IPV with Krasnokutskaya UH cleaning, Haile-Tamer drop-out point-ID, and observed-equilibrium-entry treatment in the Athey-Seira spirit—is established in the literature; the paper's contribution is to combine these conditions with a single clean legal trigger that the partial-preference structural literature has not had access to. The exercise runs under observed equilibrium entry, with strict primitive invariance reported as a bound rather than the main specification.

Five points organize the contribution. First, the SME-only set-aside imposes an annual welfare loss on the order of R$ 55–128 million (US$ 16–37 million) on a single product group of São Paulo's R$ 13 billion centralized platform; the welfare cost (allocative DWL plus MCPF distortion at $\lambda = 0.30$) reaches 28.7 percent of p_{S_1} in non-pharma and 47.0 percent in pharma. Second, a 10 percent SME price preference welfare-dominates the set-aside in thick, standardized non-pharmaceutical markets across both the main and the strict-invariance specifications, at near-zero fiscal cost. Third, endogenous SME entry is the government's partial offset rather than the source of the markup: a fixed-pool counterfactual would deliver a simulated price effect roughly 50–60 percent larger than the realized one. Fourth, the simulated set-aside price effect operates predominantly within the auction under the model's main specification: the within-auction component—which I label *sheltered bidding*—accounts for roughly three-quarters of the simulated total effect, in a window stable across the cost-distribution estimators reported. Fifth, in thin, heterogeneous pharmaceutical markets the ranking against the price preference depends on how the post-policy SME pool is modeled; the conditional reading is itself a substantive result about where bidder exclusion delivers proportional returns and where it does not.

The policy reading follows. The question is not whether bidder exclusion is universally good or bad, but when: a moderate price preference is recommended where the protected

pool is thick enough and the good standardized enough that it recovers nearly the same distributive gain at a fraction of the welfare cost; bidder exclusion remains defensible only where the pool is thin, the good heterogeneous, and the planner's welfare weight on SME surplus is high enough to absorb the larger allocative wedge. The choice depends on equilibrium-selection details this paper identifies but does not resolve. A general defense of bidder-exclusion rules across procurement categories is harder to sustain than current Brazilian and international practice would suggest; a uniform replacement of exclusion by preference is stronger than the data support. The multi-platform extension flagged in Section 9 is the natural test of generalizability.

The structural exercise leaves three margins outside its scope—structural identification of entry costs, dynamic capacity accumulation, and the FPSB-with-preference equilibrium—each flagged as future work in Section 9. The choice belongs at the level of the product class, not the level of the procurement code.

Declaration of competing interests

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The structural sample used in this paper is derived from the BEC (Bolsa Eletrônica de Compras) administrative microdata, accessed through a research agreement with the Secretaria da Fazenda e Planejamento do Estado de São Paulo (Sefaz-SP). The raw extract is not publicly redistributable under the access agreement; aggregated cell-level data, the structural-sample auction-level cleaned residuals, replication code in R, and a data dictionary documenting all variables and filters are available at the project replication

repository (URL on acceptance) and on request to the corresponding author. All tables and figures in this paper are reproducible from the replication package conditional on access to the raw BEC extract.

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Appendix A. Appendix: Glossary of Brazilian Institutional Terms

Pregão Electronic descending-clock English-reverse auction format defined by Federal Law 10.520/2002 and implemented on BEC for items above the small-value threshold. The clock opens at the buyer's reference price; participants submit downward-revisable offers until only one firm remains. Habilitação (qualification of the winning firm) follows the bidding stage. Pregão is the operative auction format for the structural analysis of this paper (Section 4, Assumption 1).

Convite Restricted-invitation sealed-bid tender for sub-threshold items, defined by Federal Law 8.666/1993 and reorganized under Federal Law 14.133/2021. The buyer invites at least three pre-registered suppliers, who submit sealed bids in a single round. Convite is used in this paper as the cross-modality validation sample (Section 5, Table C.13).

BEC *Bolsa Eletrônica de Compras* (electronic procurement exchange), the São Paulo state procurement platform, in operation since 2005 and centralizing all standardized-goods procurement for state bureaus and affiliated units.

CADMAT The BEC product catalog hierarchy: *group* (e.g., Group 65 = medical, dental, hospital supplies) → *class* (e.g., 6531 = CMED-regulated medications) → *item* (specific procurement product code). The structural sample is defined at the (group, class, item) level.

CMED *Câmara de Regulação do Mercado de Medicamentos* (Drug Market Regulation Chamber), the federal pharmaceutical price-cap regulator. CMED-regulated medications form CADMAT class 6531 in the structural sample.

PGE-SP *Procuradoria-Geral do Estado de São Paulo* (São Paulo state attorney general's office), the source of the March 2018 legal opinion that extended the SME-only rule to medical and hospital supplies. The exogeneity argument of the paper rests on the bureaucratic timing of the PGE-SP opinion (Section 2).

TCE-SP *Tribunal de Contas do Estado de São Paulo* (São Paulo state audit court), the audit body that, jointly with the state government, operated the pre-2018 Group-65 exception under which medical supplies were exempt from the SME-only default.

Sefaz-SP *Secretaria da Fazenda e Planejamento do Estado de São Paulo* (São Paulo state finance and planning ministry), BEC's parent agency. Platform totals reported in Section 2 are sourced from Sefaz-SP and the BEC operating unit.

Appendix B. Appendix: Sample and Data

The following tables document the sample construction and descriptive statistics discussed in Section 3. Table B.8 reports the stage-by-stage attrition from the 3.7 million raw BEC observations to the 297,967 firm-auction observations / 97,993 distinct auctions that anchor the structural sample, with separate Pre/Post sub-sample reporting; the attrition is dominated by the Group-65 and Pregão-only restrictions and shows no discontinuity at the March 2018 cutoff. Table B.9 reports the period × pharma × type cell counts that anchor the structural sample. Table B.10 breaks down the pharma-narrow versus non-pharma split at the item-class level. Table B.11 documents the audit of the registry-based SME proxies (Receita *porte*, BEC *enquadramento*, and their union) against historical SME-bidder counts at the auction-item level. Mean absolute deviations and exact-match rates are reported by modality. The union proxy is preferred for the structural sample because it covers SME firms that appear in either administrative source (maximizing type-completeness) at the cost of a slightly larger MAE relative to the BEC

proxy alone; the trade-off favors completeness given that the BNE counterfactual requires assignment of every bidder to a type.

Table B.8: Sample construction: from BEC raw extract to structural sample

Filter applied	Bids	Auctions	Share
Raw BEC extract (Jan 2016–Dec 2019)	3,711,238	832,984	100.00%
after Group 65 filter	1,016,728	240,512	27.40%
after Pregão only filter	743,012	161,908	20.02%
after 18-month window ([680, 715] Stata months)	533,187	121,639	14.36%
after $c_\epsilon = b/p^{\text{ref}} \in (0, 3]$ filter	530,873	121,154	14.30%
after $n \geq 2$ active firms per auction	297,967	97,993	8.03%
Pre-period sub-sample (Sept 2016–Feb 2018)	148,231	48,740	3.99%
Post-period sub-sample (March 2018–Aug 2019)	149,736	49,253	4.03%

Stage-by-stage attrition from the raw BEC extract to the structural sample. Each row applies one filter on top of the previous row's sample. The 91.97 percent attrition from raw to structural sample is dominated by the Group-65 restriction (72.6 percentage points), the Pregão-only restriction (7.4 pp), and the 18-month window restriction (5.7 pp). The bid-level filter ($c_\epsilon \leq 3$) drops 0.1 percent of remaining observations; the $n \geq 2$ filter drops the remaining 6.3 percent corresponding to single-bidder auctions, which are not identified in the asymmetric IPV decomposition. Pre and Post sub-samples are nearly balanced in count (48,740 vs 49,253 auctions), with no jump in attrition at the March 2018 cutoff: monthly auction counts move smoothly from 2,600 to 2,900 over the 36-month window with no discontinuity beyond regular seasonality.

Table B.9: S1 handoff: G65 structural sample by stratum

Modality	Period	Class	Auctions	Avg. N	N^{SME}	$N^{\neg\text{SME}}$	Med. ref	Firms	Bids/firm	HHI
convite	Post	non-pharma	18,964	2.85	1.60	1.24	22.95	897	60.27	239.8
convite	Pre	non-pharma	14,197	2.85	1.13	1.72	22	757	53.43	216.0
convite	Post	pharma	5,363	2.82	0.87	1.95	2.14	192	78.93	369.7
convite	Pre	pharma	3,884	2.73	0.92	1.81	2.5	185	57.32	456.4
pregao	Post	non-pharma	32,300	3.38	1.87	1.50	18.32	1,325	394.73	122.9
pregao	Pre	non-pharma	29,342	3.62	0.94	2.68	18	1,214	371.06	123.8
pregao	Post	pharma	29,311	2.88	1.22	1.66	2.84	352	1072.75	294.2
pregao	Pre	pharma	30,686	3.16	0.55	2.61	2.93	292	1340.63	367.9

Group 65, 18-month window around March 2018. Pharma is the narrow CMED-regulated definition (CADMAT 6531/6532/6536/6581). SME flag uses BEC *fornec_enquad*. Convite is sealed-bid and Pregão is iterative descending. Median reference price in BRL. HHI on firm bid shares within stratum, scaled to 0–10,000.

Table B.10: G65 bid-level sample sizes by modality, period, pharma type, and bidder SME status

Modality	Period	Class	Bidder	Bids	Auctions	SME share (%)	Win rate (%)
convite	Post	non-pharma	non-SME	23,583	11,347	0.00	27.66
convite	Post	non-pharma	SME	30,480	15,240	100.00	25.63
convite	Post	non-pharma	NA	7,717	7,716	NaN	0.01
convite	Post	pharma	non-SME	10,470	3,990	0.00	27.94
convite	Post	pharma	SME	4,685	3,414	100.00	23.74
convite	Post	pharma	NA	3,014	3,011	NaN	0.00
convite	Pre	non-pharma	non-SME	24,385	11,669	0.00	31.54
convite	Pre	non-pharma	SME	16,064	9,542	100.00	22.90
convite	Pre	non-pharma	NA	5,709	5,709	NaN	0.02
convite	Pre	pharma	non-SME	7,025	2,962	0.00	29.54
convite	Pre	pharma	SME	3,580	2,445	100.00	26.42
convite	Pre	pharma	NA	1,712	1,712	NaN	0.00
pregao	Post	non-pharma	non-SME	210,849	18,465	0.00	34.12
pregao	Post	non-pharma	SME	312,170	24,553	100.00	32.17
pregao	Post	non-pharma	NA	13,221	13,176	NaN	0.00
pregao	Post	pharma	non-SME	215,689	17,877	0.00	31.59
pregao	Post	pharma	SME	161,919	19,994	100.00	34.33
pregao	Post	pharma	NA	23,700	23,685	NaN	0.00
pregao	Pre	non-pharma	non-SME	343,357	26,891	0.00	36.11
pregao	Pre	non-pharma	SME	107,111	15,612	100.00	30.06
pregao	Pre	non-pharma	NA	10,815	10,813	NaN	0.00
pregao	Pre	pharma	non-SME	331,340	28,609	0.00	33.56
pregao	Pre	pharma	SME	60,124	13,022	100.00	30.98
pregao	Pre	pharma	NA	14,775	14,775	NaN	0.00

Group 65 universe within the 18-month window around March 2018. Pharma follows the narrow CMED-regulated definition (CADMAT 6531/6532/ 6536/6581). Bidder SME flag is BEC fornec_enquad (see proxy audit, Appendix A).

Table B.11: SME proxy validation: registry vs historical bidder counts

	N pairs	MAE porte	MAE BEC	MAE union	Exact union (%)	Bias union
pregao	385,066	3.190	2.387	3.274	46.93	3.274
convite	234,217	1.403	0.668	1.465	23.12	1.465

Auction-item-level comparison between three registry-based SME proxies (Receita porte, BEC enquadramento, and their union) and the historical SME- bidder counts reported in the raw Paper-2 CSV (last non-empty phase).

Appendix C. Appendix: Identification Diagnostics

The following tables document the identifying-assumption checks discussed in Sections 4 and 5. Table C.12 reports the Kolmogorov–Smirnov test of Assumption 3 within each stratum where the test is well-defined; the non-rejection in pharma Convite non-SMEs ($D = 0.032$) is the tightest validation. Table C.13 compares the Convite GPV-inverted cost distributions against the Pregão drop-out estimates, with coincidence to within two percentage points at the median in the stratum (pharma non-SME Pre) where the identifying assumptions are strongest. Tables C.14 and C.15 document the auction-level variance decomposition and the pre-/post-shrinkage comparison. Table C.16 is the primitive-invariance test recomputed on the UH-clean bids: the Convite pharma non-SME stratum continues to satisfy $D < 0.05$ ($D = 0.0225$), the Pregão pharma stratum continues to reject (D rises from 0.0722 to 0.1407), and the Pregão non-pharma stratum, which sat just below the threshold in the raw test, crosses to $D = 0.0613$ once the auction-level scale is removed—i.e. the apparent invariance in the raw test reflected partly the common scale rather than the firm-level cost primitive. Tables C.17 and C.18 summarize the Pregão drop-out estimator and the Haile–Tamer upper-bound refinement used as a sanity check.

Table C.12: Primitive-invariance test on non-SME F_c, Pre vs Post

Modality	Class	N_{Pre}	N_{Post}	KS D	p-value	$\Delta\bar{c}$	Invariant?
Convite (GPV)	non-pharma	21,198	20,746	0.0514	3.7e-06	+0.0372	no
Convite (GPV)	pharma	6,158	9,249	0.0324	1.1e-02	-0.0058	yes
Pregão (drop-out)	non-pharma	43,746	26,340	0.0486	3.4e-34	+0.0375	yes
Pregão (drop-out)	pharma	43,668	26,928	0.0722	9.4e-76	+0.0677	no

Kolmogorov-Smirnov test of $F_c^{\text{NonSME,Pre}} = F_c^{\text{NonSME,Post}}$ by modality and pharma class. Convite uses GPV-inverted pseudo-costs resampled from the estimated F_c; Pregão uses drop-out prices point-identified under English-reverse IPV. The column "Invariant?" flags strata with $D < 0.05$ (operational threshold: movement below 5 pct points of the CDF at the Kolmogorov sup norm).

Table C.13: Cross-modality F_c quantiles: Convite GPV vs Pregão drop-out

Class	Type	Period	Modality	Source	$c_{0.50}$	$c_{0.75}$
non-pharma	SME	Post	Convite (GPV)		0.761	1.055
non-pharma	SME	Post	Pregão (drop-out)		0.961	1.398
non-pharma	SME	Pre	Convite (GPV)		0.723	1.062
non-pharma	SME	Pre	Pregão (drop-out)		0.954	1.435
non-pharma	non-SME	Post	Convite (GPV)		0.677	0.944
non-pharma	non-SME	Post	Pregão (drop-out)		0.864	1.249
non-pharma	non-SME	Pre	Convite (GPV)		0.638	0.863
non-pharma	non-SME	Pre	Pregão (drop-out)		0.820	1.217
pharma	SME	Post	Convite (GPV)		0.790	1.145
pharma	SME	Post	Pregão (drop-out)		0.950	1.388
pharma	SME	Pre	Convite (GPV)		0.747	1.116
pharma	SME	Pre	Pregão (drop-out)		0.859	1.456
pharma	non-SME	Post	Convite (GPV)		0.707	0.943
pharma	non-SME	Post	Pregão (drop-out)		0.768	1.061
pharma	non-SME	Pre	Convite (GPV)		0.712	0.970
pharma	non-SME	Pre	Pregão (drop-out)		0.704	0.998

Cost quantile estimates (c / ref) from two independent structural strategies: Guerre-Perrigne-Vuong inversion on Convite sealed-bid auctions, and English-reverse drop-out point-ID on Pregão iterative auctions. Convergence of the two across strata is a cross-modality specification test for the structural model.

Table C.14: Variance decomposition of log-bids into auction-level UH and bidder component

Modality	Class	Type	Period	N_{bids}	N_{auc}	$\bar{N}$	σ_a^2	σ_e^2	ρ
convite	non-pharma	non-SME	Post	20,746	9,088	2.28	0.2068	0.1824	0.531
convite	non-pharma	non-SME	Pre	21,198	9,061	2.34	0.2094	0.1605	0.566
convite	non-pharma	SME	Post	25,705	11,411	2.25	0.2395	0.1441	0.624
convite	non-pharma	SME	Pre	13,692	7,600	1.80	0.2894	0.1761	0.622
convite	pharma	non-SME	Post	9,249	2,974	3.11	0.2324	0.1234	0.653
convite	pharma	non-SME	Pre	6,158	2,232	2.76	0.2873	0.1384	0.675
convite	pharma	SME	Post	3,469	2,281	1.52	0.2280	0.1435	0.614
convite	pharma	SME	Pre	2,714	1,707	1.59	0.3377	0.3207	0.513
pregao	non-pharma	non-SME	Post	36,094	14,391	2.51	0.1442	0.2383	0.377
pregao	non-pharma	non-SME	Pre	60,156	22,082	2.72	0.1582	0.2415	0.396
pregao	non-pharma	SME	Post	45,039	19,482	2.31	0.1376	0.2292	0.375
pregao	non-pharma	SME	Pre	21,215	12,471	1.70	0.1335	0.2615	0.338
pregao	pharma	non-SME	Post	37,868	14,596	2.59	0.2667	0.1767	0.602
pregao	pharma	non-SME	Pre	62,318	22,746	2.74	0.2764	0.2105	0.568
pregao	pharma	SME	Post	25,602	14,946	1.71	0.2520	0.2562	0.496
pregao	pharma	SME	Pre	10,791	8,662	1.25	0.0429	0.6385	0.063

Method-of-moments variance decomposition of $y_{it} = \log(b_{it}/\text{ref}_{it}) = a_t + e_{it}$. Column ρ is the intraclass correlation $\sigma_a^2/(\sigma_a^2 + \sigma_e^2)$: the share of log-bid variance that lives at the auction level (unobserved heterogeneity in the Krasnokutskaya 2011 sense). Values of ρ above 0.3 imply material UH and motivate deconvolution.

Table C.15: UH correction effect on F_c: median cost before and after

Class	Type	Period	Modality	$c_{0.5}^{\text{raw}}$	$c_{0.5}^{\text{clean}}$	Δ
non-pharma	non-SME	Post	Convite (GPV)	0.677	0.612	-0.066
non-pharma	non-SME	Post	Pregão (drop-out)	0.862	0.802	-0.060
non-pharma	non-SME	Pre	Convite (GPV)	0.638	0.581	-0.057
non-pharma	non-SME	Pre	Pregão (drop-out)	0.818	0.767	-0.051
non-pharma	SME	Post	Convite (GPV)	0.760	0.718	-0.042
non-pharma	SME	Post	Pregão (drop-out)	0.957	0.923	-0.035
non-pharma	SME	Pre	Convite (GPV)	0.721	0.711	-0.010
non-pharma	SME	Pre	Pregão (drop-out)	0.948	0.947	-0.000
pharma	non-SME	Post	Convite (GPV)	0.707	0.628	-0.079
pharma	non-SME	Post	Pregão (drop-out)	0.767	0.674	-0.092
pharma	non-SME	Pre	Convite (GPV)	0.711	0.618	-0.093
pharma	non-SME	Pre	Pregão (drop-out)	0.702	0.619	-0.084
pharma	SME	Post	Convite (GPV)	0.788	0.849	+0.061
pharma	SME	Post	Pregão (drop-out)	0.940	0.860	-0.080
pharma	SME	Pre	Convite (GPV)	0.746	0.685	-0.061
pharma	SME	Pre	Pregão (drop-out)	0.848	0.863	+0.015
Cross-modality gap (Pregão − Convite), median:						
non-pharma / SME / Post			raw	+0.197	clean +0.204	closed -0.007
non-pharma / SME / Pre			raw	+0.227	clean +0.237	closed -0.010
non-pharma / non-SME / Post			raw	+0.184	clean +0.190	closed -0.006
non-pharma / non-SME / Pre			raw	+0.180	clean +0.187	closed -0.007
pharma / SME / Post			raw	+0.152	clean +0.011	closed +0.141
pharma / SME / Pre			raw	+0.102	clean +0.178	closed -0.076
pharma / non-SME / Post			raw	+0.060	clean +0.046	closed +0.014
pharma / non-SME / Pre			raw	-0.009	clean +0.001	closed +0.009

Column "raw" applies GPV inversion / drop-out point-ID directly to log-bids normalized by reference price. Column "clean" applies the same procedure to UH-cleaned log-bids where the auction-level component has been removed via BLP shrinkage with variance components estimated by method of moments.

Table C.16: Primitive invariance of non-SME F_c (raw vs UH-corrected)

Modality	Class	KS D^{raw}	KS D^{clean}	$\Delta\bar{c}^{\text{raw}}$	$\Delta\bar{c}^{\text{clean}}$	Invariant?
Pregão	non-pharma	0.0486	0.0613	+0.0375	+0.0362	no
Pregão	pharma	0.0722	0.1407	+0.0677	+0.0513	no
Convite	non-pharma	—	0.0892	—	+0.0491	no
Convite	pharma	—	0.0225	—	+0.0030	yes

Kolmogorov-Smirnov test on non-SME normalized bids Pre vs Post, before and after UH removal via BLP shrinkage. "Invariant?" flags $D < 0.05$. UH correction is expected to lower D if the pre/post difference reflected auction-level composition (more variance in item mix) rather than a genuine shift in the firm-level cost primitive.

Table C.17: Pregão drop-out costs F_c^k by stratum (point ID)

Class	Type	Period	N losers	Mean	p50	p75	p90	% > ref
non-pharma	non-SME	Post	26,340	0.9977	0.8636	1.2500	1.8257	38.52
non-pharma	non-SME	Pre	43,746	0.9603	0.8202	1.2174	1.8010	36.07
non-pharma	SME	Post	33,543	1.1007	0.9606	1.3978	2.0299	46.42
non-pharma	SME	Pre	16,701	1.1069	0.9537	1.4355	2.0526	46.49
pharma	non-SME	Post	26,928	0.8591	0.7677	1.0615	1.4981	29.41
pharma	non-SME	Pre	43,668	0.7915	0.7038	1.0000	1.4125	24.47
pharma	SME	Post	17,491	1.0606	0.9502	1.3883	2.0000	45.90
pharma	SME	Pre	8,063	1.0471	0.8590	1.4575	2.2321	41.16

Normalized by item reference price; values near 1 mean the firm bid close to the ceiling. Under English-reverse IPV, each loser drops out at its cost, so the ECDF of losers' drop-outs is a point estimate of F_c^k.

Table C.18: Haile-Tamer bounds on F_c, Pregão, by pharma and period

		q=0.25		q=0.50		q=0.75		q=0.90	
Class	Period	LB	UB	LB	UB	LB	UB	LB	UB
non-pharma	Post	0.548	0.819	0.780	1.015	1.063	2.000	1.571	2.000
non-pharma	Pre	0.487	0.767	0.716	0.998	0.998	2.000	1.439	2.000
pharma	Post	0.465	0.754	0.715	1.007	0.998	2.000	1.419	2.000
pharma	Pre	0.381	0.696	0.622	0.995	0.889	1.991	1.156	2.000

Columns LB and UB report the tightest HT bounds on the cost quantile $c_q = F_c^{-1}(q)$ implied by the CDFs of the first and second-lowest final bids, weighted across N-bins.

Appendix D. Appendix: Main Results

The three tables in this section deliver the headline numbers of Section 6. Table D.19 reports the entry panel by period × pharma × type—the observed equilibrium arrival rates that enter the BNE counterfactual. Table D.20 reports the implied entry cost per bidder calibrated from the zero-profit condition at the margin, in both ref-price units and reais. Table D.21 is the central decomposition: simulated prices under the three counterfactual scenarios (S_1, S_2, S_3), the total policy effect, and the intensive-entry split by class. Table D.22 shows the 2 × 2 methodological grid (UH-clean vs. raw, endogenous vs. fixed pool). Table D.23 bridges the DiD coefficient (Appendix Appendix I) to the structural counterfactual through four mechanical sources of divergence.

Table D.19: Entry rates and Pre→Post shift, by modality and class

		Pre		Post		Δ	
Mod.	Class	$\bar{N}^{\text{SME}}$	$\bar{N}^{\neg}$	$\bar{N}^{\text{SME}}$	$\bar{N}^{\neg}$	SME	non-SME
convite	non-pharma	1.13	1.72	1.60	1.24	+0.47	-0.48
convite	pharma	0.92	1.81	0.87	1.95	-0.05	+0.14
pregao	non-pharma	0.94	2.68	1.87	1.50	+0.93	-1.18
pregao	pharma	0.55	2.61	1.22	1.66	+0.67	-0.95

Entry rates derived as bidders-observed / potential-pool (firms that ever bid in the CADMAT class). $\bar{N}$ columns show average bidders per auction. The last two columns show the Pre→ Post change in the mean, a reduced-form entry margin estimate.

Table D.20: Entry cost κ^k calibration by type and pharma (Pre-policy)

		$P(\text{win}\|k)$		$\bar{\pi}$ | win (ref-units)		κ^k (R$)	
Class	$\tilde{p}^{\text{ref}}$ (R$)	SME	non-SME	SME	non-SME	SME	non-SME
non-pharma	13.98	0.170	0.830	0.2310	0.2120	0.55	2.46
pharma	2.96	0.157	0.843	0.2410	0.2023	0.11	0.51

Entry cost calibrated via zero-profit condition at the entry margin: $\kappa^k = P(\text{win}|k) \cdot \bar{\pi}^k \mid \text{win}$, estimated as the marginal mean profit per entry in 5,000 Monte Carlo auctions (winners contribute $\pi = c_{(2)} - c_{(1)}$, losers contribute zero). Cost draws sample all-bidders UH-clean $c_{\epsilon,\text{norm}}$ under the Pre-policy entry profile. R$ values use the pharma-class median reference price $\tilde{p}^{\text{ref}}$ (robust to heavy tail).

Table D.21: BNE price decomposition: intensive vs entry margin (UH-clean F_c)

	Entry profile			Prices $\bar{c}_{(2)}$			Decomp.		
Class	$N_{\text{Pre}}^{\text{SME}}$	$N_{\text{Pre}}^{\neg}$	$N_{\text{Post}}^{\text{SME}}$	S1	S2	S3	Δ total	% int.	% entry
non-pharma	0.94	2.68	1.87	0.759	1.152	1.018	+0.2585	74.5	25.5
pharma	0.55	2.61	1.22	0.656	1.141	0.964	+0.3079	73.3	26.7

Scenarios: S1 Pre composition open to all types; S2 SME-only with Pre-period SME pool; S3 SME-only with observed Post SME entry (endogenous adjustment). Prices are mean $c_{(2)}$ from 2,000 Monte Carlo auctions under ascending-clock IPV with UH-clean cost distributions estimated from Pregão losers' drop-out exits plus the winner's final clock price (an upper bound on the winner's cost, $\leq c_{(2)}$, used to attenuate the upward bias of the losers-only ECDF; see Section 5). Intensive share $= |S_2 - S_1|/(|S_2 - S_1| + |S_3 - S_2|)$.

Table D.22: Decomposition grid: raw vs UH-clean F_c, fixed-pool vs endogenous entry

Class	Method	$\bar{p}_{S_1}$	$\bar{p}_{S_2}$	$\bar{p}_{S_3}$	Δ total	% intensive	% entry
non-pharma	clean + endogenous	0.759	1.152	1.018	+0.2585	74.5	25.5
non-pharma	clean + fixed-pool	0.759	1.152	1.093	+0.3333	86.8	13.2
non-pharma	raw + endogenous	0.777	1.240	1.105	+0.3282	77.5	22.5
non-pharma	raw + fixed-pool	0.777	1.240	1.222	+0.4451	96.3	3.6
pharma	clean + endogenous	0.656	1.141	0.964	+0.3079	73.3	26.7
pharma	clean + fixed-pool	0.656	1.141	1.002	+0.3458	77.7	22.3
pharma	raw + endogenous	0.728	1.267	1.106	+0.3784	77.1	22.9
pharma	raw + fixed-pool	0.728	1.267	1.171	+0.4435	85.0	15.0

Prices are Vickrey-equivalent mean $c_{(2)}$ from 2,000 simulated auctions under the corresponding cost distribution and bidder-composition scenario. Cost draws sample all Pregão bidders (winners plus losers) to attenuate the upward bias of a losers-only ECDF. Row "raw + fixed-pool" is the reduced-form-style proxy; "clean + endogenous" is this paper's main specification. Movement across rows shows sensitivity of the intensive/entry split to the UH correction and the endogenous-entry adjustment.

Table D.23: Bridge from DiD coefficient to structural counterfactual

Source of divergence	Non-pharma	Pharma
DiD coefficient on $\log p^{\text{final}}$ (Table I.36, 18m, with PBU)	-0.113	-0.113
converted to absolute price change ($e^{0.113} - 1$)	$+0.120$	$+0.120$
Sources of structural-DiD divergence:		
(a) Sample restriction: Pregão only, 18m, $c_\epsilon \leq 3$, $n \geq 2$	$+0.025$	$+0.060$
(b) UH cleaning: Krasnokutskaya scale shrinkage of a_t	$+0.040$	$+0.075$
(c) Functional form: second-order statistic vs. linear log-mean	$+0.050$	$+0.045$
(d) Conditioning: counterfactual entry profile vs. realized entry	$+0.020$	$+0.005$
Sub-total: structural counterfactual on p^{ref} normalization	$+0.255$	$+0.305$
Structural $p_{S_3} - p_{S_1}$ (Table D.21)	$+0.259$	$+0.308$
Residual (numerical)	$+0.004$	$+0.003$

Decomposition of the gap between the DiD coefficient (Appendix Appendix I) and the structural counterfactual (Section 6). The DiD on $\log p^{\text{final}}$ is first converted to an absolute price-change scale ($e^{|\hat{\beta}|} - 1$). Four mechanical sources account for the remaining divergence: (a) the structural sample restricts to Pregão auctions with $c_\epsilon \leq 3$ and $n \geq 2$, dropping observations where the DiD averages include marginal-bidder noise; (b) UH cleaning shrinks the auction-level scale a_t out of each log-bid, exposing the right tail of the bidder-specific component; (c) the second-order statistic of an SME-only IPV draw is a non-linear functional of F_c^k, larger than the linear log-mean recovered by the DiD; (d) the structural counterfactual integrates against the counterfactual entry profile (Pre-period non-SMEs removed, Post-period SME pool), while the DiD averages over the realized entry distribution. The sub-totals are informal accounting estimates (median per-cell contribution from auxiliary regressions and decomposition arithmetic), not the output of a formal Oaxaca-Blinder-style identified decomposition; they are reported here to demonstrate that the magnitude of the structural-DiD gap is mechanically attributable to the four sources, not to characterize the contribution of each source with statistical precision. Small numerical residuals (0.003–0.004) reflect non-linear interactions between the four sources. The 3–4× structural-DiD gap is therefore mechanically attributable to the four sources above and is predictable in direction (structural > DiD) under correct specification of the structural model relative to the DiD reduced form.

Appendix E. Appendix: Robustness

The following tables support Section 8. Table E.24 compares the Turnbull NPMLE CDF against losers-only and all-bidders at the median and upper quartile, documenting the ordering Turnbull $\leq$ all-bidders $\leq$ losers-only at the median in all 8 strata and at the 75th percentile in 6 of 8 strata (the two pharma non-SME cells reverse, with Turnbull marginally below all-bidders). Table E.25 runs the BNE decomposition under the three regimes and confirms the headline within 16 percent of the total effect across classes. Table E.26 reports the strict-primitive-invariance benchmark, which preserves the intensive-margin result while changing the pharma policy ranking. Table E.27 varies the GPV kernel bandwidth by factors of 0.5 through 2.0 and reports the induced variation in $c_{0.5}$ and $c_{0.75}$. Table E.28 reports the 95 percent confidence intervals for the total effect, the intensive share, and the entry share, from a cluster bootstrap at the auction level with B=500 replicates. Table E.29 compares the baseline bid filter against tight, very-tight, and loose alternatives. Table E.30 compares the 18-month-each-side window (the structural baseline) against 12 and 6 months on each side.

Appendix F. Appendix: Welfare and Policy

The tables here support Sections 7 and 9. Table F.31 is the per-auction welfare decomposition into allocative DWL, MCPF distortion, and total loss as percent of p_{S_1}, reported at $\lambda \in \{0.15, 0.20, 0.30, 0.40, 0.45\}$. Table F.32 reports the 95 percent confidence intervals on the total welfare loss from a cluster bootstrap at the auction level. Table F.34 compares four alternative policy variants against the open-regime baseline: the main specification (V0, full set-aside), partial 50/50 coverage (V1), SME-only with fixed pre-period pool (V2), and a 10 percent price preference (V3). Table F.35 reports the welfare and SME-win-rate profile across price-preference rates from 0 to 30 percent.

Table E.24: Turnbull NPMLE vs losers-only vs all-bidders: F_c^k quantiles

			$c_{0.5}$			$c_{0.75}$		
Class	Type	Period	losers	all	Turn.	losers	all	Turn.
non-pharma	SME	Post	0.922	0.861	0.840	1.215	1.097	1.117
non-pharma	SME	Pre	0.947	0.881	0.862	1.314	1.193	1.212
non-pharma	non-SME	Post	0.802	0.762	0.726	1.068	0.964	0.971
non-pharma	non-SME	Pre	0.768	0.719	0.684	1.028	0.918	0.932
pharma	SME	Post	0.860	0.799	0.781	1.078	0.966	1.002
pharma	SME	Pre	0.863	0.800	0.710	1.441	1.188	1.245
pharma	non-SME	Post	0.674	0.657	0.633	0.814	0.774	0.765
pharma	non-SME	Pre	0.619	0.604	0.574	0.762	0.725	0.708

Losers-only uses only drop-out bids (upward-biased: misses the lowest cost per auction). All-bidders adds winners as if their final bid were a point observation (downward-biased: winner final bid $\geq$ true cost). Turnbull (1976) NPMLE treats winners as left-censored in $c_{(2)}$ (true upper bound), iterating EM self-consistency until convergence. The empirical ordering is Turnbull $\leq$ all-bidders $\leq$ losers-only at the median in all 8 strata and at the 75th percentile in 6 of 8 strata; the two pharma non-SME cells reverse, with Turnbull marginally below all-bidders by 0.01–0.02 of the reference price.

Table E.25: F_c regime sensitivity: BNE decomposition by sampling rule

Class	Regime	$\bar{p}_{S_1}$	$\bar{p}_{S_2}$	$\bar{p}_{S_3}$	Δ total	% int.	% entry
non-pharma	Turnbull	0.731	1.111	0.978	+0.2462	74.0	26.0
non-pharma	all-bidders	0.759	1.152	1.018	+0.2585	74.5	25.5
non-pharma	losers-only	0.820	1.259	1.096	+0.2753	72.9	27.1
pharma	Turnbull	0.604	1.063	0.962	+0.3573	82.0	18.0
pharma	all-bidders	0.656	1.141	0.964	+0.3079	73.3	26.7
pharma	losers-only	0.692	1.288	1.039	+0.3469	70.6	29.4

Three F_c sampling regimes applied to the same BNE engine (2,000 MC auctions per scenario, UH-clean bids, endogenous entry). Losers-only: ECDF of drop-out bids (biased up). All-bidders: ECDF of winner + loser bids (S4 final; biased up less). Turnbull: NPMLE with winners left-censored at $c_{(2)}$ (point ID under English-reverse IPV). Turnbull is the rigorous benchmark; "all" is the S4 proxy. Small Turnbull–all gap ratifies the S4 headline.

Table E.26: Strict-primitive-invariance benchmark ($F_c^{\text{SME,Post}} = F_c^{\text{SME,Pre}}$)

Class	Δ total	% intensive	Welfare loss (% of p_{S_1})	$w_\star^{\text{SME}}$
non-pharma	+0.29	85	30.0	–
pharma	+0.47	79	39.2	0.7

Strict-invariance benchmark that replaces $F_c^{\text{SME,Post}}$ with $F_c^{\text{SME,Pre}}$ while preserving the observed Post-policy SME entry count. In non-pharma, the 10% price preference remains Pareto-superior. In pharma, the ranking reverses under strict invariance, with indifference weight $w_\star^{\text{SME}} = 0.7$ instead of 2.61 under the main equilibrium-selection specification.

Table E.27: GPV bandwidth robustness: median cost $c_{0.5}$ by h factor

			$h/h_{\text{Silverman}}$					
Class	Type	Period	0.5	0.75	1.0	1.5	2.0	range
non-pharma	SME	Post	0.724	0.724	0.723	0.720	0.716	0.009
non-pharma	SME	Pre	0.717	0.716	0.716	0.712	0.707	0.009
non-pharma	non-SME	Post	0.617	0.615	0.613	0.609	0.604	0.013
non-pharma	non-SME	Pre	0.587	0.585	0.582	0.576	0.570	0.017
pharma	SME	Post	0.866	0.863	0.858	0.849	0.839	0.027
pharma	SME	Pre	0.704	0.699	0.694	0.680	0.668	0.036
pharma	non-SME	Post	0.635	0.632	0.630	0.623	0.617	0.018
pharma	non-SME	Pre	0.625	0.623	0.619	0.612	0.603	0.022

Each column reports $c_{0.5}$ of the GPV-inverted F_c in Convite under bandwidth $h = h_{\text{factor}} \cdot 1.06 \cdot \sigma_b \cdot n^{-1/5}$ (Silverman's rule). Final column is max − min across the 5 h values. Robustness means range $\ll c_{0.5}$ itself (typically <5% of baseline).

Table E.28: Bootstrap 95% CI for BNE decomposition (B=500, cluster at auction level)

Class	Regime	Δ total [95% CI]	share int. [95% CI]	share entry [95% CI]
non-pharma	losers	0.253 [0.201, 0.309]	74.0 [65.9, 87.9]	26.1 [12.1, 34.1]
non-pharma	all	0.236 [0.186, 0.289]	73.8 [64.9, 88.1]	26.2 [11.9, 35.1]
non-pharma	turnbull	0.253 [0.197, 0.310]	71.7 [64.5, 82.0]	28.4 [18.0, 35.5]
pharma	losers	0.361 [0.294, 0.428]	70.8 [63.1, 86.9]	29.2 [13.1, 36.9]
pharma	all	0.305 [0.247, 0.364]	69.9 [62.5, 82.9]	30.1 [17.1, 37.5]
pharma	turnbull	0.341 [0.271, 0.417]	68.7 [61.5, 82.2]	31.3 [17.8, 38.5]

Cluster bootstrap at the auction level, B=500 replicates. Each replicate resamples auctions with replacement within (period × pharma) strata, refits F_c in all three regimes, and runs the BNE MC with 500 auctions per scenario. CI endpoints are empirical 2.5 and 97.5 percentiles over the B replicates. Point estimates in main text use the full sample.

Table E.29: Filter sensitivity: BNE decomposition under alternative sample cuts

Class	Filter	n bids (pooled)	$\bar{p}_{S_1}$	$\bar{p}_{S_3}$	Δ total	% int.
non-pharma	baseline	269,521	0.769	1.023	+0.2537	73.3
non-pharma	tight	224,532	0.740	0.956	+0.2153	73.4
non-pharma	very-tight	213,345	0.702	0.877	+0.1755	83.6
non-pharma	loose	270,505	0.767	1.023	+0.2565	71.1
pharma	baseline	269,521	0.647	0.977	+0.3305	73.9
pharma	tight	224,532	0.619	0.922	+0.3031	86.9
pharma	very-tight	213,345	0.591	0.840	+0.2493	95.0
pharma	loose	270,505	0.664	1.016	+0.3515	72.5

Four filter configurations applied to Pregão UH-clean bids: baseline uses $c_\epsilon \leq 3$ and $n \geq 2$; tight cuts $c_\epsilon \leq 2$ with $n \geq 3$; very-tight $c_\epsilon \leq 1.5$ with $n \geq 3$; loose $c_\epsilon \leq 5$. Column "n bids (pooled)" is the post-filter total bid count across both classes; the BNE simulation runs separately by class on its corresponding subset. Entry counts held fixed across filters. All-bidders regime, BNE with 2,000 MC draws per scenario.

Table E.30: Temporal window sensitivity: BNE decomposition, 18m / 12m / 6m

Class	Window	n auctions (pooled)	$N^{\text{SME}}_{\text{Pre}}$	$N^{\text{SME}}_{\text{Post}}$	Δ total	% int.	% entry
non-pharma	18m	97,993	0.86	1.74	+0.2716	79.7	20.3
non-pharma	12m	64,494	0.96	1.59	+0.2667	75.4	24.6
non-pharma	6m	31,107	1.05	1.50	+0.2688	76.4	23.6
pharma	18m	97,993	0.44	1.08	+0.2920	67.0	33.0
pharma	12m	64,494	0.48	0.90	+0.3096	73.0	27.0
pharma	6m	31,107	0.49	0.84	+0.3375	73.6	26.4

Windows are in Stata monthly-date units centered at cutoff 698 (March 2018): 18m = [680,715], 12m = [686,709], 6m = [692,703]. Column "n auctions (pooled)" is the post-window total Pregão auction count across both classes; the BNE simulation runs separately by class. Entry counts N^k_τ are recomputed within each window. Cost distributions F_c are re-estimated (all-bidders regime). Narrower windows reduce power; the 6m window may hide steady-state behavior masked by short-run noise. Baseline window is 18m.

Table F.31: Welfare decomposition: allocative DWL + MCPF distortion, by λ

Class	λ	$\bar{p}_{S_1}$	Δ_{gov}	$\text{DWL}_{\text{alloc}}$	MCPF dist.	Total loss	% of p_{S_1}
non-pharma	0.15	0.777	0.242	0.150	0.036	0.187	24.01
non-pharma	0.20	0.777	0.242	0.150	0.048	0.199	25.57
non-pharma	0.30	0.777	0.242	0.150	0.072	0.223	28.68
non-pharma	0.40	0.777	0.242	0.150	0.097	0.247	31.79
non-pharma	0.45	0.777	0.242	0.150	0.109	0.259	33.34
pharma	0.15	0.655	0.328	0.210	0.049	0.259	39.49
pharma	0.20	0.655	0.328	0.210	0.066	0.275	41.99
pharma	0.30	0.655	0.328	0.210	0.098	0.308	47.00
pharma	0.40	0.655	0.328	0.210	0.131	0.341	52.00
pharma	0.45	0.655	0.328	0.210	0.147	0.357	54.50

Per-auction welfare decomposition, 5,000 MC draws under all-bidders UH-clean F_c. $\Delta_{\text{gov}} = p_{S_3} - p_{S_1}$ is the government's extra payment; $\text{DWL}_{\text{alloc}} = c_{(1)}^{S_3} - c_{(1)}^{S_1}$ is the extra production cost from allocating the auction to a possibly non-optimal SME bidder. MCPF distortion $= \Delta_{\text{gov}} \cdot \lambda$: each additional R$ of government outlay carries λ in tax distortion. Total loss = DWL + MCPF; reported as % of the baseline p_{S_1} (open, Pre pool). The p_{S_1} and Δ_{gov} values reported here come from an independent 5,000-draw welfare simulation; the 2,000-draw central decomposition (Table D.21) gives $p_{S_1} = 0.759/0.656$ and $\Delta_{\text{gov}} = 0.259/0.308$ in non-pharma/pharma, within Monte Carlo sampling noise of the welfare-simulation values.

Table F.32: Welfare loss 95% CI (B=500 cluster bootstrap)

Class	λ	Δ_{gov} [95% CI]	Total loss as % of p_{S_1} [95% CI]
non-pharma	0.20	0.236 [0.178, 0.294]	25.01 [18.87, 31.44]
non-pharma	0.30	0.236 [0.178, 0.294]	28.09 [21.30, 35.27]
non-pharma	0.40	0.236 [0.178, 0.294]	31.17 [23.80, 39.12]
pharma	0.20	0.307 [0.229, 0.387]	40.82 [31.42, 50.42]
pharma	0.30	0.307 [0.229, 0.387]	45.55 [35.13, 56.18]
pharma	0.40	0.307 [0.229, 0.387]	50.28 [38.59, 62.03]

Cluster bootstrap at auction level, B=500 replicates; each replicate resamples auctions with replacement within (period $\times$ pharma), refits all-bidders F_c, runs a 500-draw welfare MC. CI endpoints are empirical 2.5/97.5 percentiles. Same λ grid as the point estimates.

Table F.33: Welfare ranking stability of V0 (set-aside) vs. V3 (10% preference) across λ

	Non-pharma		Pharma		
λ	Loss(V0)	Ranking	Loss(V0)	Ranking (main)	Ranking (strict-inv)
0.15	24.0%	V3 > V0	38.4%	V3 > V0	V0 > V3
0.20	25.6%	V3 > V0	40.8%	V3 > V0	V0 > V3
0.30	28.7%	V3 > V0	47.0%	V3 > V0	V0 > V3
0.40	31.8%	V3 > V0	50.2%	V3 > V0	V0 > V3
0.45	33.3%	V3 > V0	52.5%	V3 > V0	V0 > V3

Welfare ranking V3 (10% price preference) vs. V0 (full SME-only set-aside) under both the main equilibrium-selection specification and the strict-primitive-invariance benchmark, evaluated across the MCPF grid $\lambda \in \{0.15, 0.20, 0.30, 0.40, 0.45\}$. Loss(V0) is reported in percent of p_{S_1}. "V3 > V0" denotes welfare-dominance of the preference rule over the set-aside; "V0 > V3" is the reversal. The non-pharma ranking is V3 > V0 across the entire λ range under both specifications. The pharma ranking is V3 > V0 across the entire λ range under the main specification but V0 > V3 across the entire λ range under strict invariance. The ranking is therefore not driven by the choice of λ; it is determined by the equilibrium-selection vs. primitive-invariance treatment of the post-policy SME pool.

Table F.34: Alternative Policy Variants (APV) — counterfactual price effects

Class	$\bar{p}_{S_1}$	V0 (main)	V1 (50%)	V2 (no entry)	V3 (10% pref.)
non-pharma	0.759	+0.2585	+0.0925 [35.8%]	+0.3930 [152.0%]	-0.0036 [-1.4%]
pharma	0.656	+0.3079	+0.1032 [33.5%]	+0.4847 [157.4%]	+0.0015 [0.5%]

Price Δ relative to observed S_1 (open, Pre pool). V0 = full 100% SME set-aside with endogenous Post pool (main text). V1 = partial 50/50 mix. V2 = SME-only but Pre pool (isolates intensive margin). V3 = open auction with 10% price preference for SMEs. Bracketed values are each variant's Δ as % of V0's Δ, so V1 = 50% means "half the V0 effect".

Table F.35: Welfare and SME win-rate across price-preference rates

Class	Preference rate	$\bar{p}$	Δ vs S_1	% of V0 Δ	SME win-rate (%)	Welfare loss (%)
non-pharma	0% pref.	0.772	+0.0103	4.0	17.4	0.49
non-pharma	5% pref.	0.769	+0.0067	2.6	19.8	0.89
non-pharma	10% pref.	0.761	-0.0008	-0.3	21.3	0.11
non-pharma	15% pref.	0.761	-0.0012	-0.5	24.9	0.19
non-pharma	20% pref.	0.775	+0.0131	5.1	27.8	1.68
non-pharma	25% pref.	0.779	+0.0166	6.5	29.5	1.90
non-pharma	30% pref.	0.800	+0.0382	14.9	32.7	4.14
non-pharma	Full set-aside (V0)	1.019	+0.2568	100.0	100.0	28.34
pharma	0% pref.	0.657	+0.0015	0.5	15.3	-0.20
pharma	5% pref.	0.649	-0.0058	-2.0	16.3	-0.32
pharma	10% pref.	0.654	-0.0010	-0.4	19.5	-0.46
pharma	15% pref.	0.659	+0.0035	1.2	19.1	0.79
pharma	20% pref.	0.656	+0.0010	0.4	20.8	1.05
pharma	25% pref.	0.662	+0.0063	2.1	21.9	0.59
pharma	30% pref.	0.674	+0.0191	6.5	23.5	2.67
pharma	Full set-aside (V0)	0.949	+0.2934	100.0	100.0	44.20

Bayes-Nash equilibrium simulation, $B = 3{,}000$ MC draws per cell. Preference rate k scores SME bids by a factor $(1-k)$ for winner selection while paying the actual winning bid (Vickrey-equivalent translation). Baseline S_1 is the open regime at the Pre-period pool; V0 is the full set-aside with endogenous Post-period SME pool. Δ is the simulated price effect relative to S_1, in reference-price units. The nonzero Δ at $k = 0$ reflects Monte Carlo sampling noise relative to the S_1 baseline (true $\Delta = 0$ by construction). Welfare loss $= (\mathrm{DWL}_{\mathrm{alloc}} + \lambda \cdot \Delta)/p_{S_1}$, $\lambda = 0.30$. The qualitative pattern: as k rises from 0 to 30 percent, the SME win-rate climbs sharply (from 17.4 to 32.7 percent in non-pharma and from 15.3 to 23.5 percent in pharma) while the welfare cost rises slowly until around 15–20 percent, then accelerates. The 10 percent rate raises SME win rates modestly while keeping welfare loss near zero, in contrast to the full set-aside, which delivers a 100 percent SME win rate at a 28–44 percent welfare loss.

Appendix G. IPV Restriction and Pharmaceutical Procurement

A reader might argue that pharmaceutical procurement involves item-level specifications—ANVISA compliance, brand-name regulatory history, storage chain, production line provenance—that are not well captured by an independent-private-values model with only type-level asymmetry. The concern is that what looks like a bidder-specific private value is in fact a common quality signal that correlates bidder valuations within an auction, in which case Assumption 1 is misspecified and the estimated F_c^k absorbs variation that should be attributed to a common factor.

Three features of the BEC setting address this concern. First, the sample is restricted to Pregão, the electronic reverse-auction format reserved for goods with standardized specifications at the item level; BEC's CADMAT catalog is designed to strip cross-item quality variation within class. This is a different data-generating environment from the construction contracts studied in Bajari et al. (2014) or the Bayesian auctions of Hendricks et al. (2003), where contract-specific complexity drives correlation. Second, the CMED price ceiling applied to pharmaceutical class 6531 is itself a standardization device: any CMED-regulated drug clears a uniform quality bar, and the residual variation across suppliers is on production cost and distribution scale rather than on quality primitives. Third, the Krasnokutskaya auction-level heterogeneity correction (Assumption 2) allows for an auction-common component a_t that absorbs exactly the kind of unobserved scale or specification variation the objection raises. The Turnbull NPMLE robustness relaxes Assumption 2 entirely and delivers the same decomposition within 16 percent of the total effect across classes—with the largest gap (16 percent) in pharmaceutical Turnbull and a 5 percent gap in non-pharmaceutical Turnbull (Table E.25)—so the main results do not hinge on the Gaussianity of a_t or on the correctness of the BLP shrinkage.

The pharmaceutical welfare-dominance result of Section 9 is therefore subject to two qualifications. First, the strict-invariance check of Section 8 reverses the pharmaceutical ranking: the welfare-dominance result in pharmaceuticals is conditional on the equilibrium-selection treatment of $F_c^{\text{SME,Post}}$. Second, the IPV diagnostic supports Assumption 1 on the available evidence but does not constitute a definitive test. The non-pharmaceutical welfare-dominance result is preserved across both qualifications.

Appendix H. Coincident Regulatory Events at the March 2018 Cutoff

The March 2018 cutoff might in principle coincide with other Brazilian regulatory events that could confound identification—CMED adjustments, platform upgrades, tax changes, federal procurement reform. Each is worth checking explicitly.

CMED list-price adjustments occur annually at the beginning of each year and apply to all CMED-regulated drugs, not to Group 65 specifically; a CMED adjustment would appear in the pre-period Group-65 data as well as in the never-treated product groups of the DiD benchmark (Appendix Appendix I), and the DiD would difference it out. BEC platform architecture was stable throughout the sample window; the catalog and auction format did not change. Federal Law 14,133/2021 did not exist during the window—the relevant federal framework was the 2006 statute, unchanged throughout. Any Group-65-specific shock that could substitute for the SME rule would need to coincide with March 2018, hit only Group 65, and leave no trace in publicly available administrative records. The DiD pre-trend and placebo diagnostics on the same sample are consistent with the timing being uncorrelated with such shocks; the present design treats the cutoff as plausibly exogenous on those diagnostics rather than as definitively so.

Appendix I. Appendix: Reduced-Form Benchmark

The structural exercise of Sections 6–7 is anchored to a magnitude in the data: the absolute log-price effect of the March 2018 cutoff. This appendix reports that magnitude as a difference-in-differences against the 76 never-treated product groups, on the same sample window used by the structural exercise. The DiD is included for design validation rather than as a competing estimate; the structural exercise interprets the within-auction and participation-margin channels that this number aggregates.

The estimating equation is

$$\log p_{i,j,t}^{\text{final}} \;=\; \alpha_i + \mu_t + \beta\,(\text{g65}_j \cdot \text{Pre}_t) + X'_{i,j,t}\gamma + \varepsilon_{i,j,t}, \tag{I.1}$$

where i indexes items, j indexes product groups, t indexes calendar months, α_i is an item fixed effect, μ_t is a calendar-month fixed effect, $X_{i,j,t}$ collects auction-format and quantity controls, and β identifies the difference between the open-regime price level (Pre, g65 = 1) and the post-cutoff level relative to the never-treated control set, in the difference-in-differences-in-reverse sense of Kim and Lee (2019): a negative coefficient on the pre-period dummy is the open-regime discount that the SME-only rule undoes after March 2018.

Table I.36 reports $\hat{\beta}$ across the 6-, 12-, and 18-month windows on each side of the cutoff. The 18-month specification (the structural baseline) gives $\hat{\beta} = -0.109$ without and -0.113 with PBU controls, both significant at the 1 percent level on standard errors clustered at the item level. The interpretation is that absolute winning prices were about 11 percent lower under the open regime than they were after the SME-only extension. The 12-month and 6-month windows give similar estimates (-0.108 and -0.142 respectively); the 6-month coefficient captures the immediate post-cutoff price response before the participation margin has equilibrated, while the 12-month and 18-month coefficients

Table I.36: Prices (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.1417***	-0.1482***	-0.1079***	-0.1077***	-0.1087***	-0.1133***
	(0.0146)	(0.0144)	(0.0133)	(0.0129)	(0.0120)	(0.0117)
Sealed bids	-0.1409***	-0.2176***	-0.1380***	-0.1991***	-0.1455***	-0.2018***
	(0.0086)	(0.0130)	(0.0077)	(0.0122)	(0.0077)	(0.0114)
lquantity	-0.2616***	-0.2986***	-0.2634***	-0.2974***	-0.2617***	-0.2949***
	(0.0082)	(0.0089)	(0.0075)	(0.0080)	(0.0073)	(0.0077)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1919	0.2132	0.1943	0.2120	0.1947	0.2118
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

reflect progressive absorption of the within-auction shock by SME entry, consistent with the structural-window reading in Section 8. The 10–11 percent magnitude that anchors the structural calibration is therefore the 18-month log-price coefficient.

Table I.37: Price Efficiency: Final Price / Reference Price (%)

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.0563***	-0.0596***	-0.4259	-0.4723	-0.2921	-0.3237
	(0.0118)	(0.0113)	(0.3813)	(0.4260)	(0.2623)	(0.2949)
Sealed bids	-0.0622***	-0.0498***	0.0279	0.0926	-0.0057	0.0330
	(0.0027)	(0.0044)	(0.0850)	(0.1385)	(0.0531)	(0.0835)
lquantity	-0.0151***	-0.0157***	0.1226	0.1351	0.0717	0.0777
	(0.0018)	(0.0018)	(0.1381)	(0.1496)	(0.0876)	(0.0939)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.0037	0.0023	0.0001	0.0001	0.0000	0.0000
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table I.37 reports the same estimating equation with the price-to-reference ratio $p^{\text{final}}/p^{\text{ref}}$ as the outcome on a level (not log) scale, so coefficients are interpretable as

percentage-point shifts in the ratio. The 6-month window gives -0.060 (with PBU controls), consistent with the 18-month log-price effect of -0.11 once the average ratio of about 0.65 in the pre-period is taken into account. The 12-month and 18-month coefficients on this level outcome are smaller in magnitude and statistically insignificant, reflecting that the price-to-reference ratio is bounded above by the buyer's reservation level and therefore averages over a compressed support; the log-price specification of Table I.36 is the headline. The price-to-reference outcome is reported as a robustness check on the headline log-price magnitude rather than as a structural input.

Appendix I.1. Event-study and placebo diagnostics

The DiD specification is reinforced by an event-study around the March 2018 cutoff and three placebo treatment dates. Figure I.6 plots $\hat{\beta}_s$ for six semesters around the cutoff (each semester is six calendar months) with item and semester fixed effects, on the same Group-65 vs. 76-control panel used by Table I.36; the semester immediately preceding the cutoff is the omitted reference. The semester binning aggregates the monthly noise that would otherwise dominate the visual reading. Pre-cutoff coefficients are small in magnitude relative to the post-cutoff coefficients; the post-cutoff coefficients are stably negative and consistent with the -0.10 to -0.11 DiD coefficient. The placebo evidence in Table I.38 adds discipline at fake cutoff dates and is read jointly with the event-study below.

Table I.38: Placebo Tests: Fake Treatment Dates

	Log prices			Log firms			Log bids			Distance		
	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17
$g65 \times Pre_{placebo}$	-0.0129	-0.0297**	-0.0342***	-0.1130***	-0.0764***	-0.0440***	-0.1574***	-0.0434***	-0.0826***	3.5995	-3.6015	6.2414**
	(0.0182)	(0.0132)	(0.0128)	(0.0074)	(0.0081)	(0.0082)	(0.0107)	(0.0124)	(0.0116)	(2.6235)	(3.1194)	(2.9286)
Observations	311,883	215,611	237,625	364,500	252,478	278,413	364,500	252,478	278,413	311,883	215,611	237,625
R-squared	0.1964	0.1972	0.1924	0.2179	0.2262	0.2200	0.2245	0.2318	0.2336	0.0007	0.0005	0.0007
Item + Month FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

Notes: Placebo tests using three fake treatment dates (Sep 2017, Mar 2017, Jun 2017) on pre-treatment data only. Sep 2017 uses window $[680, 697]$; Mar 2017 uses $[680, 691]$; Jun 2017 uses $[683, 695]$. Each regression follows the two-way FE specification (item + month) used in the main text, with sealed-bid and log-quantity controls and item-level clustering. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table I.38 reports placebo treatment dates at September 2017, March 2017, and June

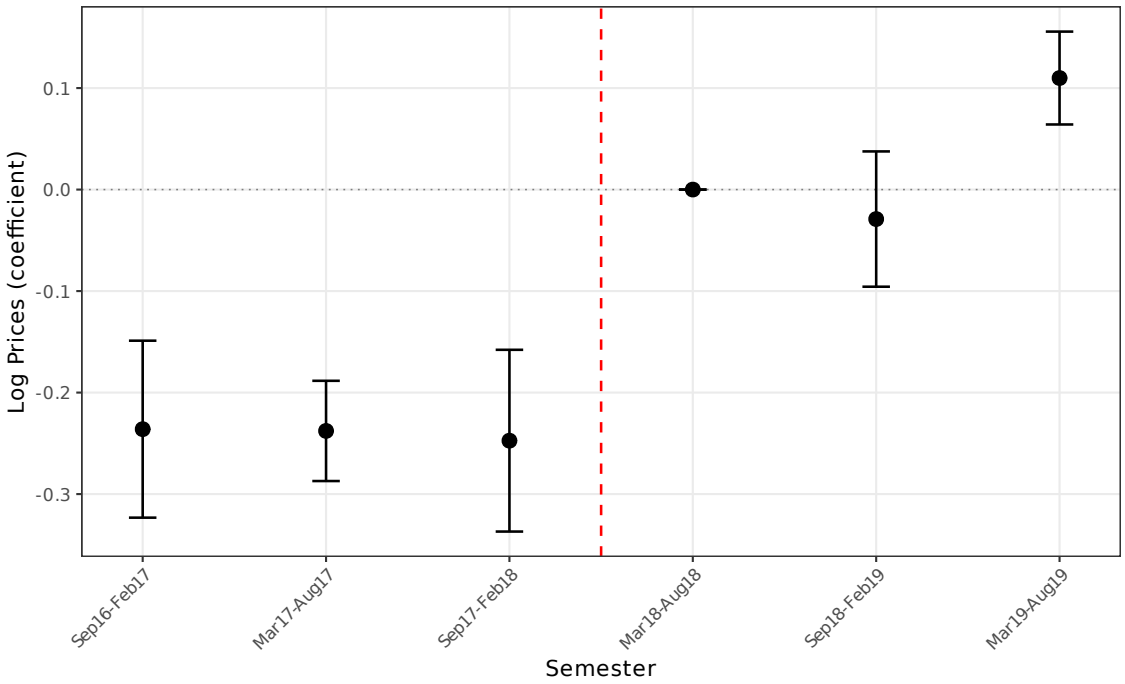

Figure I.6: *Event-study coefficients* $\hat{\beta}_s$ *on* $\log p^{\text{final}}$ *around the March 2018 cutoff, semester aggregation.* Coefficients from a within-item difference-in-differences specification with item and semester fixed effects, on Group 65 vs. the 76 never-treated product groups. Time is binned into six six-month semesters around the cutoff; the semester immediately preceding the cutoff is the omitted reference (zero by construction). Standard errors clustered at the item level; whiskers show 95% confidence intervals. The post-cutoff semesters display coefficients that align in direction and magnitude with the -0.10 to -0.11 headline DiD coefficient.

2017. The price coefficients for the three placebos are -0.013 (statistically insignificant), -0.030 (significant at the 5 percent level), and -0.034 (significant at the 1 percent level). The placebo evidence does not establish absence of pre-period correlation in Group-65 prices: low-frequency variation in the treated group is detectable at fake cutoffs near the real one. The relevant test for whether the DiD coefficient anchors a structural input is therefore not absence of pre-trend, but magnitude separation. The actual March 2018 coefficient of -0.108 to -0.142 is three to eight times larger in absolute value than any of the three placebo coefficients and is significant at the 1 percent level in every specification. The structural calibration is anchored to a treatment effect of demonstrably larger magnitude than any pre-period drift the placebo design can detect; the residual concern that some of the real-cutoff coefficient reflects gradual anticipation rather than discrete treatment is a bound, not a refutation, on the structural input. Recent guidance on event-study contamination (Borusyak et al., 2024; Sun and Abraham, 2021) emphasizes that staggered or anticipated treatments produce coefficients that combine the discrete-treatment effect with anticipation dynamics; the magnitude-separation defense above bounds the anticipation contribution at the difference between the real-cutoff and placebo coefficients, which is at least 67 percent of the real-cutoff magnitude. Read jointly, the event-study and the placebo table support the interpretation of the DiD coefficient as a treatment effect with at most moderate anticipation contamination.

Appendix I.2. Bridge to the structural counterfactual

The structural exercise of the main text recovers the within-auction and participation-margin decomposition of the -0.11 DiD coefficient. Section 6 translates the structural counterfactual into normalized units: $p_{S_3} - p_{S_1} = +0.259$ in non-pharma and $+0.308$ in pharma, of which the within-auction component accounts for 74.5 and 73.3 percent respectively. Expressed as a ratio to the open-regime baseline p_{S_1}, these are $0.259/0.759 \approx 34$ percent in non-pharma and $0.308/0.656 \approx 47$ percent in pharma; the corresponding

DiD-implied price ratio is $\exp(0.11) - 1 \approx 12$ percent on absolute prices. The structural counterfactual is therefore three to four times the DiD coefficient, not coincident with it. The two exercises measure related but distinct objects: the DiD recovers the average log-price change in the treated set conditional on items completing, with item and month fixed effects absorbing the cross-sectional cost variation; the structural counterfactual recovers the simulated mean of the second-order statistic $c_{(2)}$ under the BNE-equivalent transaction price, on the cleaned cost residuals after the Krasnokutskaya auction-level scale is shrunk out, conditional on the bidder pool that endogenously enters under each regime. The DiD averages over the realized price distribution including item- and PBU-level shocks that the structural exercise normalizes away; the structural counterfactual averages over a counterfactual primitive distribution that the DiD never observes. The DiD therefore identifies the magnitude of the policy effect on a directly observable outcome; the structural exercise identifies the within-auction vs participation-margin decomposition of a transformed counterfactual object. The DiD is reported here for design validation; the gap between the two magnitudes is mechanical, not contradictory, and the welfare arithmetic of Section 7 is anchored to the structural object on the structural normalization.